

Office
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**GOVERNMENT OF ANDHRA PRADESH
WATER RESOURCES DEPARTMENT**

**Name of Work:- "Construction of double lane bridge at Km.18.390
on Budameru drain in Vijayawada city limits of
in Krishna district under Modernization of krishna
delta drainage system"**

**Estimate Amount :-Rs:354.70 Lakhs
SSR 2018-19**

**OFFICE OF THE SUPERINTENDING ENGINEER
IRRIGATION CIRCLE,
VIJAYAWADA**

Office of the Chief Engineer,
Krishna Delta System, Vijayawada.

Proc No: CE/KDS/VJA/OT-4/AEE-11/Drains Krishna/ 368 Dated: 5-02-2019:

Sub:- Water Resources Department - Irrigation - "Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta System" - Technical Sanction - Accorded - Regarding

- Ref:- 1. GO.Rt.No.592 Water Resources (Project-II) Department, Dt:21.08.2018
2. CE /CDO/VJA Lr. No. CE/CDO/EE/Div6/DEE2/KDS/DLRB @KM18.390 on Budameru Dt:1-2-2019
3. SE IC VJA Lr.No.SE/IC/VJA/DB/ATO.2/42^{CE}, Dt:04.02.2019.

Under the circumstances reported by the Superintending Engineer vide reference 3rd cited, the estimate for the work of "Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta System" for an amount of Rs. 354.70 Lakhs ((Rupees Three Crores Fifty Four Lakhs And Seventy Thousand Only) is technically sanctioned and registered vide CE/KDS R.No: 26 CE/ 2018-19, Dt:5-02-2019 and copy of Technical sanctioned Estimate is herewith communicated for information and taking further necessary action.

The Expenditure is debitable to the head of account "MH 4700-Capital outlay on Major Irrigation - 01 Major Irrigation- Commercial M.H.136 Krishna Delta System G.H.11 State Development Plan S.H. (27) Canals and Distributaries 530 Major works-531 Other Expenditure".

Encl: Copy of Abstract of Tec. Sanctioned Estimate.

To
The Superintending Engineer,
Irrigation Circle, Vijayawada.

Sd/- R.Sathish Kumar 5-2-2019
Chief Engineer
Krishna Delta System
Vijayawada.

//t.c.f//
M. Tuleganneti 5/2/19
Officer Technical-4
K10
5/2/19

CHECK SLIP ACCOMPANYING THE PROJECT / SCHEME / WORK ESTIMATE

1	Name of the Work	:	Name of work: Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta Drainage System
2	Estimate Amount Rs.	:	354 70 000
3	Category of Project / Scheme / Work	:	budameru River --Budameru Basin Bridge
4	Location / District / Mandal / Village	:	in between Velagaleru Regulator and Enikepadu UT
5	Ayacut proposed - Wet / ID	:	Budameru River--Budameru Basin
6	Cropping pattern	:	
7	Source / River basin / Sub-Basin / Sub-Minor basin	:	Not applicable
8	Availability of water at the site	:	Budameru River --Budameru Basin
9	Allocation of water / utilization	:	Not applicable
10	Reference in which Hydrological clearance was accorded by the competent authority	:	Not applicable
11	Cost per Acre	:	Not applicable
12	B.C. Ratio	:	Not applicable
13	Whether the report accompanying the estimate, detailed estimate and abstract	:	Not applicable
14	Whether salient features are enclosed	:	Yes
15	Whether command area plan / Index plan enclosed	:	Yes
16	Whether the approved Designs / Drawings / Hydraulic particulars / Cross sections are	:	Index map enclosed
17	Whether C.C. lining thickness Grade of concrete and CNS layer thickness proposed are with reference to the relevant IS codes.	:	Yes
18	Whether the data enclosed is based on the current SSR with year	:	Not applicable
19	Whether latest cement / steel rates are adopted	:	Yes
20	Whether the quotations for non SSR items are enclosed	:	Yes
21	Whether the lead statement with certificates is enclosed	:	Not applicable
22	Whether barrow area / quarry maps are enclosed	:	Yes

23	Whether the amount of labour component and certificate of LA & LI furnished	:	Not applicable
24	Whether geological and foundation investigation carried out	:	Not applicable
25	Whether the extent of LA / forest lands is furnished.	:	Not applicable
26	Are the rates of L.A. adopted based on the certificate issued by the Revenue authorities.	:	Not applicable
27	Whether provision for compensatory afforestation and NPV is provided in consultation with forest dept.	:	Not applicable
28	Whether the cost for CM & CD works is based on cost curves updated with current SSR	:	Not applicable
29	Whether provision for distributories and field channels is based on the estimate prepared for a model block covering 10% of the	:	Not applicable
30	Whether provisions made for dewatering and other LS provisions are reasonable.	:	Not applicable
31	Whether the provision for formation of road, NH crossing, railway crossing and other road crossing is based on their specification and	:	Yes
32	Whether the site was inspected by the competent authority before submitting the estimate as indicated below. If so furnish the date of inspection and confirm whether the	:	Not applicable
33	Up to Rs. 10.00 Lakhs EE	:	
i)	Rs. 10.00 Lakhs to Rs. 50.00 Lakhs SE	:	
ii)	Above 50.00 Lakhs ENC / CE	:	
iii)	Are the detailed estimates / DATAs checked in SEs office	:	
34	Are the provisions examined / abstract estimate checked in CE's office	:	
35	Whether the check slip for Admn. Approval is enclosed	:	Not applicable
36	Whether all officers up to CE signed	:	Not applicable
37	Construction programme of the Project / Scheme / Work	:	Not applicable
38	Q.C. Check authority proposed for this work	:	The work is proposed to be taken up budameru modernization.
39	Q.C. Check authority proposed for this work	:	Executive Engineer, Quality Control Division, Vijayawada


Asst Executive Engineer
Special Section
Vijayawada


Dy Executive Engineer
Special Sub Division
Vijayawada


Executive Engineer
Water Resources Department
Special Division, Vijayawada

Report accompanying the estimate for "Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta Drainage System"

**Est. Rs. 354.70 Lakhs
SSR 2018-19**

Budameru river consists of two arms, the Western arm rises in the range of hills near Zannalovu Durgam of A.Konduru Mandal and Eastern arm rises in the hill range near Annaraopet(V) of Reddigudem(M) of Krishna District. Both the arms ie. Western arm (Budameru) and Eastern arm (Kothula vagu) will join at 1.00 Km South of Nandigama(V) of G.Koduru Mandal. Pulivagu another hill stream joining Budameru at 2.00 Kms upstream side of Budameru diversion Regulator at Velagaleru(V) limits. The river flows southwards for about 8.50 Km upto Rayanapadu(V) and turns east and runs about 28 Kms and crosses Eluru canal through Erikepadu under tunnel and further travels to a length of 55 Km in delta area and empties into Kolleru lake. Many Major and minor tributaries join the river during its course. The total length of the river is about 125 Kms.

Government in the G.O.Ms.No.259, dt.29.11.2007 has accorded administrative approval for Rs.4573.00 Crores for Modernisation of Krishna Delta drainage system in which improvements and modernisation of Budameru and also included

The Hon'ble M.L.A, Vijayawada Central constituency has represented to the Hon'ble Minister, Water Resources, Govt of Andhrapradesh to construct a Double lane Bridge at KM.18.390 on Budameru in the Vijayawada Municipal Corporation limits in Krishna district duly calling tenders separately by meeting the expenditure from the sanctioned modernisation funds. The Surrounding area is developing soon,after the announcement of the Capital near Vijayawada City, the areas fast expanding and new colonies are forming rapidly. .

Government has accorded permission to take up the work under Non EPC under Modernisation of Krishna Delta Drainage System vide Govt. Memo No.14079/Major/VII.2/2010, dt.3.9.2010 for 7nos of DLR Bridges in different packages.

The Government in The G.O.RT No.592 Dated.21-08-2018 has accorded the permission to **"Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta Drainage System"** for Rs. 354.70 Lakhs Hve

The drawings for the bridge at KM.18.390 were approved by the Chief Central Designs Organization, Vijayawada vide drawing No: Budameru/99/2018/20/2019, Date: 01-02-2019 and as per the drawing the estimate is prepared and same will be submitted for necessary accord the permission for Technical sanctioned from the competent authority

Now the estimate is prepared with the current SSR 2018-2019 and the Estimate is worked to Rs.354.70 Lakhs.

The following L.S. Provisions are made.

1. Provision towards seigniorage charges
2. Provision towards GST charges at 12%
3. Provision towards conducting site survey and soil testing and obtaining detailed drawings from CE, CDO.
4. Provision towards SBC and conducting plate load test.
5. Provision towards NAC @0.1%.
6. Provision towards Stationery, Xerox and photographic charges.

The expenditure is debitable to Head of Account "4700 Capital out lay on Major Irrigation(01) Major Irrigation(commercial)-MH 136 Krishna Delta system-GH(11)- State Development Scheme-SH(27) Canal and distributaries-530 Major works-531 Other Expenditure".

A. K. C.
21/2/19
Asst Executive Engineer
Special Section
Vijayawada

[Signature]
Dy Executive Engineer
Special Sub Division
Vijayawada

[Signature]
Executive Engineer
Special Division
Vijayawada.

[Signature]
Superintending Engineer
IRRIGATION CIRCLE-
VIJAYAWADA
[Signature]
8-2-2020

ABSTRACT ESTIMATE

Name of work: Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta Drainage System .

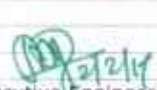
WITH 2018-19 SSR RATES

Sl. No	Quantity	Description of work	Rate	Per	Amount
Bridge Portion					
1	13094.00 Cum	Earthwork in excavation for structures as per drawing and technical specifications Clause 305.1 including setting out, construction of shoring and bracing, removal of stumps and other deleterious material and disposal upto a lead of 50 m, dressing of sides and bottom and backfilling in trenches with excavated suitable material as per Technical Specification 305 MORD / 304 MORTH (Upto 3 M depth)	101.30	1 Cum	1326422
2	6407.00 Cum	Earthwork in excavation for structures as per drawing and technical specifications Clause 305.1 including setting out, construction of shoring and bracing, removal of stumps and other deleterious material and disposal upto a lead of 50 m, dressing of sides and bottom and backfilling in trenches with excavated suitable material as per Technical Specification 305 MORD / 304 MORTH (3 M to 6 M)	116.00	1 Cum	743212
3	496.00 Cum	Filling in foundation trenches with sand as per drawing and technical specification Clause 305.3.9 MORD & 304 MORTH	514.30	1 Cum	255093
4	600.00 Hours	Dewatering by installing 10HP diesel engine including cost of diesel lubricants, oils, crew and convey engine, with accessories to where even necessary today up the site including to complete the work as directed by the Department	343.00	1 Hour	205800
5	50.00 Cum	Providing and laying insitu vibrated M-10 ^{M-15} (28 days cube compressive strength not less than 10 N / sq mm) grade cement concrete using 40 mm down size approved, clean, hard graded aggregates for foundation filling including cost of all materials, machinery, labour, formwork, cleaning, batching, mixing, placing in position, levelling, vibrating, finishing, curing etc., complete with initial lead upto 50 m and all lifts. for Levelling course	5418.00	1 Cum	270900
6	844.00 Cum	Reinforced Cement Concrete M 30 grade for footing raft including reinforcement and formwork complete as per drawing and technical specifications Clauses 800 and 1211 MORD / 1500, 1600, 1700 & 2704 MORTH. R.C.C. grade M 30 Design Mix, using 20 mm Graded Metal (MORTH 12.8 G) RBR-FNDN-4 for Raft Footing	5381.30	1 Cum	4541817
7	132.00 Cum	R.C.C. grade M 20 (Design Mix) using 20 mm Graded Metal (MORD & MORTH 12.8 C) mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days complete as per drawings for Cutoff (RCC M20) (Design Mix) using 20 mm Graded Metal (MORTH) for Cut off	5128.20	1 Cum	676922

Sl. No	Quantity	Description of work	Rate	Per	Amount
8	47.32 Cum	R.C.C. grade M 25 (Design Mix) using 20 mm Graded Metal (MORD & MORTH 12.8 E) mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days complete as per drawings for wings footing	4888.00	1 Cum	231300
9	385.00 Cum	Plain/reinforced cement concrete in substructure complete as per drawings and technical specification Clauses 802, 804, 805, 806, 807, 1202 and 1204 MORD and 1500, 1700 & 2200 MORTH for height upto 5M (piers and abutment) -RCC Grade M25 (Design mix)	5666.00	1 Cum	2181410
10	93.00 Cum	Plain/reinforced cement concrete in substructure complete as per drawings and technical specification Clauses 802, 804, 805, 806, 807, 1202 and 1204 MORD and 1500, 1700 & 2200 MORTH for height from 5M to 10M (piers) -RCC Grade M25 (Design mix)	5835.00	1 Cum	542655
11	970.00 Sqm	Providing and constructing 30 cm thick dry rubble stone pitching including cost of all materials, labour, hand packing, finishing etc., complete(rubble stones : 0.33 cum/sqm)	509.00	1Sqm	493730
12	146810.00 Kgs	Providing, fabricating and placing in position reinforcement steel bars for RCC works including cleaning, straightening, cutting, bending, hooking, lapping, welding wherever required,tying with 1.25 mm dia soft annealed steel wire, including cost of all materials, machinery, labour etc.,complete with initial lead upto 50 and all lifts..	66.00	1 Kgs	9689460
13	107.00 Cum	Providing and laying M-30 grade design mix using 20MM graded metal for reinforced cement concrete in superstructure as per drawing and technical specifications Clauses 800, 1205.4 and 1205.5 MORD / Sections 1500, 1600 & 1700 MORTH for approach slab and kerbs - RCC grade M-30.	5311.00	1 Cum	568277
14	45.00 Cum	Providing and laying cement concrete wearing course M 30 grade including reinforcement complete as per drawing and technical specifications Clauses 800 and 1206.3 MORD and 2702 MORTH for wearing coat	5456.00	1 Cum	245520
15	345.50 Cum	Providing and laying insitu vibrated M-20 (28 days cube compressive strength not less than 20 N / sq mm) grade cement concrete using 20 mm down size approved, clean, hard, graded aggregates for deck slab & kerb including cost of all materials,machinery, labour, formwork,scaffolding, cleaning, batching, mixing, placing in position, levelling, vibrating, finishing, curing etc., complete with initial lead upto 50 m and all lifts for deck slab	9278.50	1 Cum	3205722
16	123.60 Rmt	Construction of R.C.C. railing of M 25 grade in cast-in-situ with 20 mm nominal size aggregate, true to line and grade, tolerance of vertical railing post not to exceed 1 in 500, centre-to-centre spacing between vertical posts not to exceed 2000 mm as per drawing and technical specifications Clauses 800, 900 and 1208.3 MORD. For R.c.C Railing	4582.00	1 Rmt	566335

Sl. No	Quantity		Description of work	Rate	Per	Amount
17	230.67	Nos.	Providing weep holes in brick masonry/stone masonry, plain/reinforced concrete abutment, wing wall, return wall with 100 mm dia AC pipe extending through the full width of the structures with slope of 1(V):20(H) towards drawing face complete as per drawing and technical specification Clauses 614, 709, 1204.3.7 MORD / 2706 & 2200 MORTH.	576.00	1 No.	132866
18	20.00	Nos.	Drainage spouts complete as per drawing and technical specifications Clause 1209 MORD & 2705 MORTH	5791.20	1 No.	115824
19	50.70	Rmt	Providing and fixing in position 20 mm thick premoulded joint filler in expansion joint for fixed ends of simply supported spans not exceeding 10 m to cater for a horizontal movement upto 20 mm, covered with sealant complete as per drawing and technical specifications MORTH	267.00	1 Rmt	13537
20	40.56	Sqm	BMS-W-21 Mastic Pad 2' 0"x4' 0" (0.60M x 1.20 M) of 1" (25.4 MM) thick	839.00	1 Sqm	34030
21	1499.00	Cum	Providing gravel backing and laying filter material underneath pitching in slopes complete as per drawing and technical specifications Clause 1302 MORD / 2504 MORTH	1096.00	1 Cum	1642904
22	787.50	Cum	Construction of granular sub-base by providing coarse graded material, spreading in uniform layers with motor grader on prepared surface, mixing by mix in place method with rotavator at OMC, and compacting with vibratory roller to achieve the desired density, complete as per clause 401 MORTH for diversion road.	1689.00	1 Cum	1330088
23	60	Rmt	Supply of NP3 1000mm dia pipes including collars cost and conveyance of materials to site of work for diversion road.	5426.76	1 Rmt	325606
24	15	Nos.	Cost and supply of RCC collars suitable for 1000MM dia NP-3 clause pipes.	947.00	1 No.	14205
25	15.00	Nos.	Laying and jointing 1000 mm dia. NP-3 class or IRC standard hume pipes in CM 1 : 2 proportion by volume including cost of all materials (excluding pipes and collars), machinery, labour, aligning, packing joints with hemp, finishing, curing etc., complete with initial lead upto 50 m and all lifts for diversion road.	1031.00	1 No.	15465
26	255.00	Cum	Providing, Laying, Spreading and compacting graded HBG crushed stone aggregate to Wet Mix macadem specification including cost, selgnorage of all materials and including premixing the material with water at OMC in Mechanical mix plant carriage of mixed material by tipper to site, laying in uniform layers with paver in base courses on well prepared surface and compacting with Vibratory roller to acheive the desired density etc., as directed by the Engineer-in-Charge and as per MoRT&H specification.406 (4th revision) for finished item of work.	1833.00	Cum	467415

Sl. No	Quantity	Description of work	Rate	Per	Amount
27	1275.00 Sqm	Providing and laying surface dressing in single coat using 6 mm nominal size IRC HBG machine crushed stone aggregates @ 0.004 cum / sqm on a layer of bitumen binder of 80/100 grade @ 0.75 kg / sqm including cost, seigniorage and conveyance of all materials to work site and laid on prepared surface and rolling with 8-10 T Power Road Roller etc., complete for finished item of work as per MoRT&H Specification 510 (4th Revision) and as directed by the Engineer-in-charge	41.00	Sqm	52275
28	2550.00 Sqm	Providing and applying tack coat with bitumen Emulsion (Medium setting) (Bulk) using Emulsion pressure distributor at the rate of 0.20 kgs per sqm on the prepared bituminous/granular surface cleaned with mechanical broom for finished item of work as per MoRT&H Specification 503 (4th revision) and as directed by the Engineer-in-Charge.	12.00	Sqm	30600

Sl. No	Quantity	Description of work	Rate	Per	Amount
29	64.00 Cum	Providing and laying of 50mm thick Bituminous Macadam with hot mix plant producing an average output of 37.5 tonnes per hour using hard blasted granite crushed aggregates of Grading - II as per table 500-4 of MoRT&H specification 504 (4th Revision) premixed with bituminous binder of 80/100 grade @ 3.3% by weight of total mixture, transported to site, laid over a previously prepared surface with mechanical paver finisher to the required grade, level and alignment and rolled as per Clauses 501.6 and 501.7 to achieve the desired compaction for finished item of work as directed by the Engineer-in-Charge including hire and operational charges all T&P and all other contingent charges necessary including cost of seigniorage charges on all materials etc., complete and as per MoRTH specification No. 504(4th Revision)	6256.00	Cum	400384
30	32.00 Cum	Providing 25 mm thick compacted Semi-Dense Bituminous Concrete by hot mix plant using hard blasted granite crushed aggregates of Grading -2 as per table 500-15 of specification 508 of MoRT&H (4th Revision), premixed with bituminous binder 60/70 grade @ 5% of mix and filler, transporting the hot mix to work site, laying with mechanical paver finisher to the required grade, level and alignment, rolling with smooth wheeled, vibratory and tandem rollers to achieve the desired compaction as per MoRT&H Specification 508 (4th Revision) complete for finished item of work in all respects as directed by Engineer-in-charge.	7673.00	Cum	245536
31	1417.52 Sqm	Painting to new walls with 2 coats of water proof cement paint (Snowcem) of approved brand and shade over a base coat of approved cement primer grade I making making 3 coats in all to give an even shade after thoroughly brushing the surface to remove all dirt and remains of loose powdered materials, including cost and conveyance of all materials to work site and all operational, incidental, labour charges etc. complete for finished item of work as per SS 912 for internal walls	222.00	Sqm	314689
Working Items Total in Rupees					30879999
Total ECV					30879999
L.S. Items					
32 LS		Provision towards seigniorage charges			406776
33 LS		Provision towards GST @ 12%			3705600
34 LS		Provision towards conducting site survey, drilling of Bore holes and Soil testing and obtaining detailed drawings from CE, CDO.			100000
35 LS		Provision towards S.B.C and conducting plate load test			325000
36 LS		Provision towards NAC @0.1%			30880
37 LS		Provision towards photographic charges			21745
Total Amount in Rupees					35470000
(Rupees Three Crores Fifty Four Lakhs and Seventy Thousand only)					
 Asst Exe Engineer Special Section Vijayawada  Dy. Executive Engineer Special Sub Division Vijayawada  Executive Engineer Special Division Vijayawada  Superintending Engineer IRRIGATION CIRCLE VIJAYAWADA					

87.2020

Technical Sanction is hereby accorded for the work of
"Construction of Double Lane Road Bridge at km 18.390 on Budameru
Drain in Vijayawada city limits in Krishna district under Modernization
of Krishna Delta system" for an amount of Rs. 354.70 lakhs (Rupees Three
crores Fifty Four lakhs and Seventy Thousands only) and registered vide
CE/KDS/ R. NO. 26 CE/2018-19 Dt. 5/2/2019.

The Expenditure is debitables to the head of account:-
"M.H - 4700 - capital outlay on Major Irrigation - 01 Major Irrigation -
commercial M.H. 136 Krishna delta system G.H - 11 State Development plan
S.H (27) canals & distributaries - 530 Major works - 531 other expenditure".

Sl. No	Quantity	Description of work	Rate	Per	
29	64.00 Cum	Providing and laying of 50mm thick Bituminous Macadam with hot mix plant producing an average output of 37.5 tonnes per hour using hard blasted granite crushed aggregates of Grading - II as per table 500-4 of MoRTH specification 504 (4th Revision) premixed with bituminous binder of 80/100 grade @ 3.3% by weight of total mixture, transported to site, laid over a previously prepared surface with mechanical paver finisher to the required grade, level and alignment and rolled as per Clauses 501.6 and 501.7 to achieve the desired compaction for finished item of work as directed by the Engineer-in-Charge including hire and operational charges all T&P and all other contingent charges necessary including cost of seigniorage charges on all materials etc., complete and as per MoRTH specification No. 504(4th Revision)	6256.00	Cum	400384
30	32.00 Cum	Providing 25 mm thick compacted Semi-Dense Bituminous Concrete by hot mix plant using hard blasted granite crushed aggregates of Grading -2 as per table 500-15 of specification 508 of MoRTH (4th Revision), premixed with bituminous binder 60/70 grade @ 5% of mix and filler, transporting the hot mix to work site, laying with mechanical paver finisher to the required grade, level and alignment, rolling with smooth wheeled, vibratory and tandem rollers to achieve the desired compaction as per MoRTH Specification 508 (4th Revision) complete for finished item of work in all respects as directed by Engineer-in-charge.	7673.00	Cum	245536
31	1417.52 Sqm	Painting to new walls with 2 coats of water proof cement paint (Snowcem) of approved brand and shade over a base coat of approved cement primer grade I making making 3 coats in all to give an even shade after thoroughly brushing the surface to remove all dirt and remains of loose powdered materials, including cost and conveyance of all materials to work site and all operational, incidental, labour charges etc, complete for finished item of work as per SS 912 for internal walls	222.00	Sqm	314689
Working Items Total in Rupees					30879999
Total ECV					30879999
L.S. Items					
32 L.S		Provision towards seigniorage charges			406776
33 L.S		Provision towards GST @ 12%			3705600
34 L.S		Provision towards conducting site survey, drilling Bore holes and Soil testing and obtaining detailed drawings from CE, CDO			100000
35 L.S		Provision towards S.B.C and conducting plate load test			325000
36 L.S		Provision towards NAC @0.1%			30880
37 L.S		Provision towards photographic charges			21745
Total Amount in Rupees					35470000
(Rupees Three Crores Fifty Four Lakhs and Seventy Thousand only)					
Technical Sanction is hereby accorded for the work of "Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta System" for an amount of Rs. 354.70 Lakhs (Rupees Three Crores Fifty Four Lakhs and Seventy Thousand only) and registered vide CE/KDS R.No : 26 CE/2018 Dt:5/2/2019 The Expenditure is debitable to the Head of account "MH 4700- Capital outlay on Major Irrigation-01 Major Irrigation- Commercial M.H. 136 Krishna Delta System G.H- State Development Plan SH (27) Canals and Distributaries 530 Major Works-531 other expenditure"					
Sd/- R Satish Kumar, 5.9.2019 Chief Engineer Krishna Delta System Vijayawada M. Tulsaneth 5/2/19 Officer Technical-4					

S+8
 5.5

DETAILED ESTIMATE									
Name of work: Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta Drainage System .									
S.N o	Description of Work etc particulars	Measurements						Contents of	
		No			L	B	D	Area	
1	Earthwork in excavation for structures as per drawing and technical specifications Clause 305.1 including setting out, construction of shoring and bracing, removal of stumps and other deleterious material and disposal upto a lead of 50 m, dressing of sides and bottom and backfilling in trenches with excavated suitable material as per Technical Specification 305 MORD / 304 MORTH upto 3.00M Depth.								
	for Raft (avg GL +17.00mt)	1	x	1	(67+76)/2	(18.7+27.70)/2	3.000	4976.400	
	For cutoff	1	x	2	58.00	0.500	1.800	104.400	
	Short side	1	x	2	9.700	0.500	1.800	17.460	
	Drain approaches U/S & D/S	1	x	2	12.00	(56.50+65.60)/2	3.000	4395.600	
	For drain diversion	1	x	1	100.00	12.000	3.000	3600.000	
	Total quantity							13094.000	Cum
2	Earthwork in excavation for structures as per drawing and technical specifications Clause 305.1 including setting out, construction of shoring and bracing, removal of stumps and other deleterious material and disposal upto a lead of 50 m, dressing of sides and bottom and backfilling in trenches with excavated suitable material as per Technical Specification 305 MORD / 304 MORTH 3.00 to 6.00M Depth.								
	Upto 3 M depth to 6M								
	for Raft	1	x	1	(58.00+67.00)/2	(9.70+18.70)/2	3.000	2662.500	
	Drain approaches U/S & D/S	1	x	2	12.00	(47.50+56.50)/2	3.000	3744.000	
	Total quantity							6407.000	Cum
3	Filling in foundation trenches with sand as per drawing and technical specification Clause 305.3.9 MORD & 304 MORTH								
	for Raft	1	X	1	57.00	8.700	1.000	495.900	Cum
								496.000	Cum
4	Dewatering by installing 10HP diesel engine including cost of diesel lubricants, oils, crew and convey engine, with accessories to where even necessary today up the site including to complete the work as directed by the Department								
	Considering 30 day at 10 Hours/day = 30x10 Hoursx2 Nos. (10HP)	1	X	1	30.00	10.00	2.000	600.000	hrs
5	Providing concrete for plain/reinforced concrete in open foundations using 40 mm nominal size Graded hard stone aggregate, mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days complete as per drawings and technical specifications Clause 802, 803, 1202 & 1203 MORD and 1500, 1700 & 2100 MORTH.								
	For levelling course under Raft	1	X	1	57.00	8.70	0.100	50.000	Cum
								50.000	Cum
6	Reinforced Cement Concrete M 30 grade for footing raft including reinforcement and formwork complete as per drawing and technical specifications Clauses 800 and 1211 MORD / 1500, 1600, 1700 & 2704 MORTH. R.C.C. grade M 30 Design Mix, using 20 mm Graded Metal (MORTH 12.8 G) RBR-FNDN-4								
	For Raft foundation	1	X	1	58.00	9.70	1.500	844.000	Cum
7	R.C.C. grade M 20 (Design Mix) using 20 mm Graded Metal (MORD & MORTH 12.8 C) mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days complete as per drawings for Cutoff (RCC M20) (Design Mix) using 20 mm Graded Metal (MORTH)								
	All round cutoff for raft	1	X	2	58.000	0.500	1.800	104.400	
	short direction for raft	1	X	2	9.300	0.500	1.800	16.740	
	for wing wall	2	X	2	3.500	0.500	1.500	10.500	
								132.000	Cum
8	R.C.C. grade M 25 (Design Mix) using 20 mm Graded Metal (MORD & MORTH 12.8 E) mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days complete as per drawings for wings								
	for wings footing	1	X	2	8.450	3.500	0.800	47.320	Cum
9	Plain/reinforced cement concrete in substructure complete as per drawings and technical specification Clauses 802, 804, 805, 806, 807, 1202 and 1204 MORD and 1500, 1700 & 2200 MORTH for height upto 5M (piers) -RCC Grade M25 (Design mix)								
	For Abutments	1	X	2	8.450	(1.80+1.10)/2	5.000	122.525	
	For piers rectangular portion	1	X	4	7.500	1.200	5.000	180.000	
	For semicircular portion	2	X	4	3.143	0.6*0.6/2	5.000	22.629	
	wings	2	X	2	3.500	(1.2+0.5)/2	5.000	59.500	
	Total Quantity							385.000	Cum

S.N o	Description of Work etc particulars	Measurements					Contents of Area		
		No		L	B	D			
10	Plain/reinforced cement concrete in substructure complete as per drawings and technical specification Clauses 802, 804, 805, 806, 807, 1202 and 1204 MORD and 1500, 1700 & 2200 MORTH for height from 5M to 10M (piers) -RCC Grade M25 (Design mix)								
	For Abutments	1	X	2	8.450	(1.80+1.10)/2	0.702	17.203	
	For Dirt walls on abutments	1	X	2	8.450	0.500	0.790	6.676	
	For Bed Block on abutments(Cap)	1	X	2	8.450	1.100	0.300	5.577	
	For piers semi circular portion	2	X	4	3.143	0.6*0.6*/2	0.702	3.177	
	For piers rectangular portion	1	X	4	7.500	1.200	0.702	25.272	
	Bed blocks on piers(Cap)	1	X	4	8.700	1.200	0.300	12.528	
	wings	2	X	2	3.500	(1.2+0.5)/2	1.867	22.217	
	Total Quantity							93.000	Cum
11	Providing and constructing 30 cm thick dry rubble stone pitching including cost of all materials, labour, hand packing, finishing etc., complete(rubble stones : 0.33 cum/sqm)								
	300mm thick bed pitching (u/s and d/s)	1	X	2	(54+43)/2	10.000		970.000	Sqm
12	Providing, fabricating and placing in position reinforcement steel bars for RCC works including cleaning, straightening, cutting, bending, hooking, lapping, welding wherever required,tying with 1.25 mm dia soft annealed steel wire, including cost of all materials, machinery, labour etc.,complete with initial lead upto 50 and all lifts..								
	As per Statement enclosed							146810.00	Kgs
13	Providing and laying reinforced cement concrete in superstructure as per drawing and technical specifications Clauses 800, 1205.4 and 1205.5 MORD / Sections 1500, 1600 & 1700 MORTH for approach slab and kerbs.Providing and laying M-30 grade design mix using 20MM graded metal for reinforced cement concrete in superstructure as per drawing and technical specifications Clauses 800, 1205.4 and 1205.5 MORD / Sections 1500, 1600 & 1700 MORTH for approach slab and kerbs - RCC grade M-30.								
	Approach slab	1	X	2	3.500	8.450	0.300	17.745	
	Kerbs	1	X	5	10.760	0.500	0.300	8.070	
	for Kerbs of approaches road	2	X	2	65.000	0.475	0.275	33.963	
	above Kerb	2	X	2	65.000	0.300	0.600	46.800	
	Total Quantity							107.000	Cum
14	Providing and laying cement concrete wearing course M 30 grade including reinforcement complete as per drawing and technical specifications Clauses 800 and								
	For wearing coats on deck slab	1	X	1	53.800	7.500	0.100	40.350	
	For wearing coats on approach slab	1	X	2	3.500	8.450	0.080	4.732	
	Total Quantity							45.000	Cum
15	Providing and laying reinforced cement concrete in superstructure as per drawing and technical specifications Clauses 800, 1205.4 and 1205.5 MORD / Sections 1500, 1600 & 1700 MORTH for Deck slab.								
	for deck slab at 69.10 Cum/span (As per drawing MOST)	1	X	5	69.100			345.500	Cum
16	Construction of R.C.C. railing of M 20 grade in cast-in-situ with 20 mm nominal size aggregate, true to line and grade, tolerance of vertical railing post not to exceed 1 in 500, centre-to-centre spacing between vertical posts not to exceed 2000 mm as per drawing and technical specifications Clauses 800, 900 and 1208.3 MORD. For R.C.C Railing								
	for bridge portion including approach slab portion	1	X	2	61.800			123.600	Cum
17	Providing weep holes in brick masonry/stone masonry, plain/reinforced concrete abutment, wing wall, return wall with 100 mm dia AC pipe extending through the full width of the structures with slope of 1(V):20(H) towards drawing face complete as per drawing and technical specification Clauses 614, 709, 1204.3.7 MORD / 2706 & 2200 MORTH.								
	For Abutments	2	X	3	4.000			24.000	
	For wings	2	X	3	4.000			24.000	
	for approach side retaining wall at 1.5m C/C	2	X	2	68.5/1.5			182.667	
	Total Quantity							230.667	Nos
18	Drainage spouts complete as per drawing and technical specifications Clause 1209 MORD & 2705 MORTH								
		2	X	10	1.000			20.000	Nos

S.No	Description of Work etc particulars	Measurements				Contents of Area			
		No	L	B	D				
19	Providing and fixing in position 20 mm thick premoulded joint filler in expansion joint for fixed ends of simply supported spans not exceeding 10 m to cater for a horizontal movement upto 20 mm, covered with sealant complete as per drawing and technical specifications MORTH								
		1	X	6		8.450		50.700	Rmt
20	BMS-W-21 Mastic Pad 2' 0"x4' 0" (0.60M x 1.20 M) of 1" (25.4 MM) thick								
		1	X	6		8.450	0.800	40.560	Sqm
21	Providing Gravel backing and laying filter material underneath pitching in slopes complete as per drawing and technical specifications Clause 1302 MORD / 2504 MORTH								
	Gravel base for approach slab	1	X	2	3.500	8.500	0.225	13.388	
	Gravel for line side slopes	2	X	2	(10.00+16.00)/2	8.000	0.750	312.000	
	for approaches	1	X	2	45.000	8.500	(1.5+0.3)/2	688.500	
	Gravel for under 300mm thick bed pitching U/s & D/s	1	X	2	10.000	(54+43)/2	0.500	485.000	
	Total Quantity							1499.000	Cum
22	Construction of granular sub-base by providing coarse graded material, spreading in uniform layers with motor grader on prepared surface, mixing by mix in place method with rotavator at OMC, and compacting with vibratory roller to achieve the desired density, complete as per clause 401 MORTH.								
	Bridge approach approach both side	1	X	2	70.000	7.500	0.750	787.500	Cum
23	Supply of NP3 1000mm dia pipes including collars cost and conveyance of materials to site of work								
	For diversion Road	4	X	6	2.500			60.000	Rmt
24	Cost and supply of RCC collars suitable for 1000MM dia NP-3 clause pipes.								
		1	X	15				15.000	Nos
25	Laying and jointing 1000 mm dia. NP- 3 class or IRC standard hume pipes in CM 1 : 2 proportion by volume including cost of all materials (excluding pipes and collars), machinery, labour, aligning, packing joints with hemp, finishing, curing etc., complete with initial lead upto 50 m and all lifts.								
	For jointing	1	X	15				15.000	Nos
26	Providing, Laying, Spreading and compacting graded HBG crushed stone aggregate to Wet Mix macadem specification including cost,seignorage of all materials and including premixing the material with water at OMC in Mechanical mix plant carriage of mixed material by tipper to site, laying in uniform layers with paver in base courses on well prepared surface and compacting with Vibratory roller to acheive the desired density etc., as directed by the Engineer-in-Charge and as per MoRT&H specification.406 (4th revision) for finished item of work.								
	for bridge approach on both sides	1	x	2	85.000	7.500	0.200	255.000	Cum
27	Providing and laying surface dressing in single coat using 6 mm nominal size IRC HBG machine crushed stone aggregates @ 0.004 cum / sqm on a layer of bitumen binder of 80/100 grade @ 0.75 kg / sqm including cost, seignorage and conveyance of all materials to work site and laid on prepared surface and rolling with 8-10 T Power Road Roller etc., complete for finished item of work as per MoRT&H Specification 510 (4th Revision) and as directed by the Engineer-in-charge								
	for bridge approach on both sides	1	x	2	85.000	7.500		1275.000	SqM
28	Providing and applying tack coat with bitumen Emulsion (Medium setting) (Bulk) using Emulsion pressure distributor at the rate of 0.20 kgs per sqm on the prepared bituminous/granular surface cleaned with mechanical broom for finished item of work as per MoRT&H Specification 503 (4th revision) and as directed by the Engineer-in-Charge.								
	Below Bituminous Macadam	1	x	2	85.000	7.500		1275.000	
	above Bituminous Macadam	1	x	2	85.000	7.500		1275.000	
						Total Qty		2550.000	SqM
29	Providing and laying of 50mm thick Bituminous Macadam with hot mix plant producing an average output of 37.5 tonnes per hour using hard blasted granite crushed aggregates of Grading - II as per table 500-4 of MoRT&H specifaicaion 504 (4th Revision) premixed with bituminous binder of 80/100 grade @ 3.3% by weight of total mixture, transported to site, laid over a previously prepared surface with mechanical paver finisher to the required grade, level and alignment and rolled as per Clauses 501.6 and 501.7 to achieve the desired compaction for finished item of work as directed by the Engineer-in-Charge including hire and opetional charges all T&P and all other contingent charges necessary including cost of seignorage charges on all materials etc., complete and as per MoRTH specification No. 504(4th Revision)								
	for bridge approach on both sides	1	x	2	85.000	7.500	0.050	64.000	Cum

S.No	Description of Work etc particulars	Measurements						Contents of Area	
		No		L	B	D			
30	Providing 25 mm thick compacted Semi-Dense Bituminous Concrete by hot mix plant using hard blasted granite crushed aggregates of Grading -2 as per table 500-15 of specification 508 of MoRT&H (4th Revision), premixed with bituminous binder 60/70 grade @ 5% of mix and filler, transporting the hot mix to work site, laying with mechanical paver finisher to the required grade, level and alignment, rolling with smooth wheeled, vibratory and tandem rollers to achieve the desired compaction as per MoRT&H Specification 508 (4th Revision) complete for finished item of work in all respects as directed by Engineer-in-charge.								
	for bridge approach on both sides	1	x	2	85.000	7.500	0.025	32.000	Cum
31	Painting to new walls with 2 coats of water proof cement paint (Snowcem) of approved brand and shade over a base coat of approved cement primer grade I making making 3 coats in all to give an even shade after thoroughly brushing the surface to remove all dirt and remains of loose powdered materials, including cost and conveyance of all materials to work site and all operational, incidental, labour charges etc. complete for finished item of work as per SS 912 for internal walls								
	for Kerbs sides	2	x	2	61.800	0.275		67.980	
	for kerbs top	1	x	2	61.800	0.450		55.620	
	for hand rails	2	x	2	61.800	1.100		271.920	
	Approach Kerb								
	for kerb sides	4	x	2	65.000	0.275		143.000	
	for kerb top	4	x	1	65.000	0.150		39.000	
	for parapet wall sides	4	x	2	65.000	0.600		312.000	
	for parapet wall top	4	x	1	65.000	0.300		78.000	
	for retaining wall faces (straight portion)	4	x	1	5.000	2.500		50.000	
	retaining wall faces (sloped portion)	4	x	1	80.000	1.250		400.000	
					Total Qty			1417.520	Sqm

NICE
Assistant Executive Engineer
Special Section, Vijayawada

2/2/19
Deputy Executive Engineer,
Special Sub Division,
Vijayawada.

2/2/19
Executive Engineer
Special Division
Vijayawad

DATA (WITH SSR 2018-19)

Name of work: Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta Drainage System .

AS PER R&B STANDARD DATA

S. No	Description	Unit	Quantity	Rate Rs.	Amount Rs.
1	2	3	4	5	6
	RBR-FNDN-1 page.No.484				
1	Excavation for Structures				
	Earthwork in excavation for structures as per drawing and technical specifications Clause 305.1 including setting out, construction of shoring and bracing, removal of stumps and other deleterious material and disposal upto a lead of 50 m, dressing of sides and bottom and backfilling in trenches with excavated suitable material as per Technical Specification 305 MORD / 304 MORTH				
	(B) Mechanical Means				
	(i) Upto 3 m depth				
	Unit = cum				
	Taking output = 240 cum				
	a) Labour				
	Mate	day	-		
	Mazdoor (Unskilled) (Page No-8)	day	8.32	400.00	3328.00
	Add 40% area allowance on labour (C&D)				1331.20
					4659.20
	b) Machinery				
	Hydraulic Excavator 1 cum bucket capacity (Page No-27)	hour	6.00	2791.60	16749.60
	Total A+B =				21408.80
	c&d) Overheads & Contractors Profit at 13.615% on (A+B)				2914.81
	Total cost for	240	Cum		24323.61
	Rate per cum = (a+b+c+d)/240=				101.35
				or say	101.30
2	RBR-FNDN-1 page.No.484				
	Earthwork in excavation for structures as per drawing and technical specifications Clause 305.1 including				
	ii 3 m to 6 m depth				
	Unit = cum				
	B Mechanical Means				
	Taking output = 210 cum				
	a) Labour				
	Mate	day	-		
	Mazdoor (Unskilled)	day	8.32	400.00	3328.00
	Add 40% area allowance on labour (Page No -7 of SSR)				1331.20
				Total	4659.20
	b) Machinery				
	Hydraulic excavator 1.0 cum bucket capacity	hour	6.00	2791.60	16749.60
	Total A+B =				21408.80
	c&d) Overheads & Contractors Profit at 13.615% on (A+B)				2914.81
	Total cost for	210	Cum		24323.61
	Rate per cum = (a+b+c+d)/210=				115.83
				or say	116.00
3	RBR-FNDN-1 page.No.485				
	Earthwork in excavation for structures as per drawing and technical specifications Clause 305.1 including setting out, construction of shoring and bracing, removal of stumps and other deleterious material and disposal upto a lead of 50 m, dressing of sides and bottom and backfilling in trenches with excavated suitable material as per Technical Specification 305 MORD / 304 MORTH				
	III Depth above 6.0 M				
	B Mechanical Means				
	Unit = cum				
	Taking output = 180 cum				
	a) Labour				
	Mate	day	-		
	Mazdoor (Unskilled)	day	10.40	400.00	4160.00
	Add 40% area allowance on labour (Page No-7 of SSR)				1664.00
					5824.00
	b) Machinery				
	Hydraulic excavator 1.0 cum bucket capacity	hour	6.00	2791.60	16749.60

S. No	Description	Unit	Quantity	Rate Rs.	Amount Rs.
1	2	3	4	5	6
	c&d) Overheads & Contractors Profit at 13.615% on (A+B)				22573.60
	Cost for 180 cum = a+b+c+d				3073.40
	Rate per cum = (a+b+c+d)/180=				25647.00
				or say	142.48
					142.50
	RBR-FNDN-2 page.No.488				
4	Filling in foundation trenches with sand as per drawing and technical specification Clause 305.3.9 MORD & 304 MORTH				
I	Sand filling				
	Unit = cum				
	a) Labour				
	Mate	day	-		
	Mazdoor (Unskilled)	day	0.31	400.00	124.00
	Add 40% area allowance on labour				49.60
					173.60
	b) Material				
	Sand	cum	1.00	100.00	100.00
					273.60
	c&d) Overheads & Contractors Profit ON (A+B) at 13.615%				37.25
	e) Conveyance charges				
	Sand	cum	1.00	203.40	203.40
	Rate per cum = a+b+c+d				514.25
	Rate per 1 Cum =				514.30
5	Dewatering by installing 10HP diesel engine including cost of diesel lubricants, oils, crew and convey engine, with accessories to where even necessary today up the site including to complete the work as directed by the Department				
	Dewatering pump 10HP (diesel) (Page 43 of SSR)	Hour	1	343.00	343.00
	(Rate as per hire charges of plants and machinery for the year 2018-19)				
6	RBR-FNDN-4				
	Reinforced Cement Concrete M 30 grade for footing raft including reinforcement and formwork complete as per drawing and technical specifications Clauses 800 and 1211 MORD / 1500, 1600, 1700 & 2704 MORTH. C.C. grade M 30 Design Mix, using 20 mm Graded Metal (MORTH 12.8 G) RBR-FNDN-4 for Raft Footing				
	Case 1 : Using Concrete Mixer				
	Unit = cum				
	(a) Material				
	Cement	t	0.407	3900.00	1587.30
	Coarse sand	cum	0.45	100.00	45.00
	20 mm aggregate	cum	0.54	1303.00	703.62
	10 mm aggregate	cum	0.36	894.00	321.84
					2657.76
	(b) Labour				
	Mate	day	-		
	Mason (1st Class)	day	0.1	475.00	47.50
	Mazdoor (Unskilled)	day	1.39	400.00	556.00
	Add 40% area allowance on labour				603.50
					241.40
					844.90
	(c) Machinery				
	Concrete mixer 0.4/0.28 cum capacity	hour	0.40	525.50	210.20
	Generator 33 KVA	hour	0.40	1032.00	412.80
					623.00
	(a+b+c)				4125.66
	(d) Formwork @ 3.5% on (a+b+c)		3.50%		144.40
					4270.06
	e&f) Overheads & Contractors Profit		13.615%		581.37
	Rate per cum = a+b+c+d+e+f				4851.43
	e) Conveyance charges				
	Coarse sand	cum	0.45	203.40	91.53
	20 mm aggregate	cum	0.54	487.00	262.98
	10 mm aggregate	cum	0.36	487.00	175.32
	total cost for Conveyance charges				529.83
	Rate per cum =				5381.26

S. No	Description	Unit	Quantity	Rate Rs.	Amount Rs.
1	2	3	4	5	6
				say	5381.30
7	RBR-FNDN-4-C I				
	R.C.C. grade M 20 (Design Mix) using 20 mm Graded Metal (MORD & MORTH 12.8 C) mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days complete as per drawings for Cutoff (RCC M20) (Design Mix) using 20 mm Graded Metal (MORTH) for Cut off				
	CASE I : Using Concrete Mixer				
	Unit = cum				
	(a) Material				
	Cement	t	0.347	3900.00	1353.30
	Coarse sand	cum	0.45	100.00	45.00
	20 mm aggregate	cum	0.54	1303.00	703.62
	10 mm aggregate	cum	0.36	894.00	321.84
					2423.76
	(b) Labour				
	Mate	day	-		
	Mason (1st Class)	day	0.1	475.00	47.50
	Mazdoor (Unskilled)	day	1.39	400.00	556.00
					603.50
	Add 40% area allowance on labour				241.40
					844.90
	(c) Machinery				
	Concrete mixer 0.4/0.28 cum capacity	hr / cum	0.4	525.50	210.20
	Generator 33 KVA	hour	0.4	1032.00	412.80
					623.00
	(a+b+c)				3891.66
	(d) Formwork @ 4% on (a+b+c)		4.00%		155.67
					4047.33
	e&f) Overheads & Contractors Profit		13.615%		551.04
	Rate per cum = a+b+c+d+e+f				4598.37
	g)Conveyance charges				
	Coarse sand	cum	0.45	203.40	91.53
	20 mm aggregate	cum	0.54	487.00	262.98
	10 mm aggregate	cum	0.36	487.00	175.32
	Total conveyance charges				529.83
	Rate per cum				5128.20
				say	5128.20
8	RBR-FNDN-4-E page.No.493				
	R.C.C. grade M 25 (Design Mix) using 20 mm Graded Metal (MORD & MORTH 12.8 E) mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days complete as per drawings for Raft and foundation for Wings footing				
	Case 1 : Using Concrete Mixer				
	Unit = cum				
	(a) Material				
	Cement	t	0.403	3900.00	1571.70
	Coarse sand (Screened)	cum	0.45	100.00	45.00
	20 mm aggregate	cum	0.54	1303.00	703.62
	10 mm aggregate	cum	0.36	894.00	321.84
					2642.16
	(b) Labour				
	Mate	day	-		
	Mason (1st Class)	day	0.1	475.00	47.50
	Mazdoor (Unskilled)	day	1.39	400.00	556.00
					603.50
	Add 40% area allowance on labour				241.40
					844.90
	(c) Machinery				
	Concrete mixer 0.4/0.28 cum capacity	hr / cum	0.4	525.50	210.20
					3697.26
	(d) Formwork @ 3.75% on (a+b+c)		3.75%		138.65
					3835.91
	e&f) Overheads & Contractors Profit at 13.615% on (a+b+c+d)				522.26
	Rate per cum = a+b+c+d+e+f				4358.17
	g)Conveyance charges				
	Coarse sand (Screened)	cum	0.45	203.40	91.53
	20 mm aggregate	cum	0.54	487.00	262.98

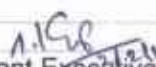
S. No	Description	Unit	Quantity	Rate Rs.	Amount Rs.
1	2	3	4	5	6
	10 mm aggregate	cum	0.36	487.00	175.32
	Total conveyance charges				529.83
	Rate per cum				4888.00
				say	4888.00
9	RBR-SBST-5 VI(i) page.No.560 Plain/reinforced cement concrete in substructure complete as per drawings and technical specification Clauses 802, 804, 805, 806, 807, 1202 and 1204 MORD and 1500, 1700 & 2200 MORTH for height upto 5M (piers) -RCC Grade M25 (Design mix)				
	VI R.C.C. grade M 25 (Design Mix)				
	Unit = cum				
	(i) Height upto 5m				
	Case I : Using concrete Mixer				
	(a) Material				
	Cement	t	0.403	3900.00	1571.70
	Coarse sand (Screened)	cum	0.450	100.00	45.00
	20 mm aggregate	cum	0.540	1303.00	703.62
	10 mm aggregate	cum	0.350	894.00	321.84
					2642.16
	(b) Labour				
	Mate	day	-		
	Mason (1st Class)	day	0.10	475.00	47.50
	Mazdoor (Unskilled)	day	1.39	400.00	556.00
					603.50
	Add 40% area allowance on labour				241.40
					844.90
	(c) Machinery				
	Concrete mixer 0.4/0.28 cum capacity	hour	0.40	525.50	210.20
	Generator 33 KVA	hour	0.40	1032.00	412.80
	Vibrator	hour	0.00	0.00	0.00
					623.00
					4110.06
	(d) Formwork @ 10% on (a+b+c)		10.00%		411.01
					4521.07
	e&f) Overheads & Contractors Profit @13.615% on (a+b+c+d)				615.543
					5136.61
	g)Conveyance charges				
	Coarse sand (Screened)	cum	0.450	203.40	91.53
	20 mm aggregate	cum	0.540	487.00	262.98
	10 mm aggregate	cum	0.360	487.00	175.32
					529.83
	Rate per cum				5666.44
				say	5666.00
10	RBR-SBST-5 VI(ii) page No.561 Plain/reinforced cement concrete in substructure complete as per drawings and technical specification Clauses 802, 804, 805, 806, 807, 1202 and 1204 MORD and 1500, 1700 & 2200 MORTH for height upto 5M (piers) -RCC Grade M25 (Design mix)				
	(ii) Height 5m to 10m				
	Case I : Using concrete Mixer				
	Per Cum Basic Cost of Labour, Material & Machinery (a+b+c) of Item VI (i) Case I excluding of Formwork	cum	1		
	a+b+c=				4110.06
	d) formwork				
	Add on cost of material, labour and machinery (a+b+c) for Formwork		11.80%		484.99
	Add on cost of material, Labour and machinery excluding formwork to cater for extra lift		1.80%		73.98
					4669.03
	e&f) Overheads & Contractors Profit @13.615% on(a+b+c+d)				635.69
	Rate perm (a+b+c+d+e+f)				5304.72
	g)Conveyance charges				
	Coarse sand (Screened)	cum	0.450	203.40	91.53
	20 mm aggregate	cum	0.540	487.00	262.98
	10 mm aggregate	cum	0.360	487.00	175.32
					529.83
	Rate per Cum				5834.55
					5835.00

S. No 1	Description 2	Unit 3	Quantity 4	Rate Rs. 5	Amount Rs. 6
11	i RBR-SPST-13 .ii, page.594				
	Reinforced Cement Concrete M 30 grade approach slab including reinforcement and formwork complete as per drawing and technical specifications Clauses 800 and 1211 MORD / 1500, 1600, 1700 & 2704 MORTH				
	Cement concrete M30 Grade Refer relevant item of concrete in item 1(III) by using batching plant, excluding formwork i.e. per cum basic cost (a+b+c) (Excluding OH & CP) (Page 577 (iii))				
	(a) Material				
	Cement	t	0.407	3900.00	1587.30
	Coarse sand (Screened)	cum	0.45	100.00	45.00
	20 mm aggregate	cum	0.54	1303.00	703.62
	10 mm aggregate	cum	0.36	894.00	321.84
	ii				2657.76
	(b) Labour				
	Mate	day	-		
	Mason (1st Class)	day	0.10	475.00	47.50
	Mazdoor (Unskilled)	day	1.39	400.00	556.00
					603.50
	Add 40% on corporation area allowance.				241.40
					844.90
	c) Machinery				
	Concrete mixer 0.4/0.28 cum capacity	hour	0.40	525.50	210.20
	Generator 33 KVA	hour	0.40	1032.00	412.80
					623.00
	(a+b+c) =				4125.66
	Formwork may be added at the rate of 2 percent Cum on (a+b+c)				82.51
					4208.17
	e&f) Overheads & Contractors Profit at 13.615% on (a+b+c)				572.94
					4781.12
	g)Conveyance charges				
	Coarse sand (Screened)	cum	0.45	203.40	91.53
	20 mm aggregate	cum	0.54	487.00	262.98
	10 mm aggregate	cum	0.36	487.00	175.32
					529.83
	Rate per Cum				5310.95
				say	5311.00
12	(i) RBR-SPST-5 (i) Page No.588				
	Providing and laying cement concrete wearing course M 30 grade including reinforcement complete as per drawing and technical specifications Clauses 800 and 1206.3 MORD and 2702 MORTH for wearing coat				
	Unit = cum				
	(a) Material				
	Cement	t	0.407	3900.00	1587.30
	Sand	cum	0.45	100.00	45.00
	20 mm aggregate	cum	0.54	1303.00	703.62
	10 mm aggregate	cum	0.36	894.00	321.84
	HYSD bar reinforcement including binding wire (Rate as per item 13.2) except OH & CP	t	0.075		0.00
					2657.76
	(b) Labour				
	Mate	day	-		
	Mason (1st Class)	day	0.10	475.00	47.50
	Mazdoor (Unskilled)	day	1.39	400.00	556.00
	Mazdoor (Unskilled) for cleaning deck slab concrete surface	day	0.15	400.00	60.00
					663.50
	Add 40% area allowance on labour				265.40
					928.90
	c) Machinery				
	Concrete mixer 0.4/0.28 cum capacity	hour	0.40	525.50	210.20
	Generator 33 KVA	hour	0.40	1032.00	412.80
					623.00
	(a+b+c)				4209.66
	(d) Formwork @ 3% of cost of concrete		3.00%		126.29
					4335.95
	e&f) Overheads & Contractors Profit @ 13.615%				590.34
					4926.29

S. No 1	Description 2	Unit 3	Quantity 4	Rate Rs. 5	Amount Rs. 6
	g)Conveyance charges				
	Sand (Screened)	cum	0.45	203.40	91.53
	20 mm aggregate	cum	0.54	487.00	262.98
	10 mm aggregate	cum	0.36	487.00	175.32
					529.83
	Rate per cum			or say	5456.12
				say	5456.00
13	I				
	RBR-SPST-7(i) Page No.590				
	Construction of R.C.C. railing of M 25 grade in cast-in-situ with 20 mm nominal size aggregate, true to line and grade, tolerance of vertical railing post not to exceed 1 in 500, centre-to-centre spacing between vertical posts not to exceed 2000 mm as per drawing and technical specifications Clauses 800, 900 and 1208.3 MORD. For R.C.C Railing				
	Unit = Running m				
	Taking output = 4x12 m				
	Span = 48 m				
	(a) M 25 grade R.C.C.				
	No. of vertical posts = (6+1) 4 = 28 nos				
	Cross-sectional area of vertical post = 0.25x0.275 = 0.069 sqm				
	Concrete in vertical posts = 0.069x28x1.00 = 1.932 cum				
	Hand rail in 3 tiers = 3x48 = 144 m				
	Cross-sectional area = 0.17x0.175 = 0.03 sqm				
	Concrete in hand rails = 0.03 x 144 = 4.32 cum				
	Total concrete = 1.932+4.32 = 6.252 cum	cum	6.252	4581.76	28645.18
	(b) HYSD bar reinforcement (Rate as per item 13.2)	t	1.36	43500.00	59160.00
					87805.18
	c&d) Overheads & Contractors Profit @ 13.615%				11954.68
	Cost for 48 m = (a+b+c+d)	48 M			99759.86
	g)Add Conveyance charges				
	Coarse sand	cum	0.45	203.40	91.53
	20 mm aggregate	cum	0.54	487.00	262.98
	10 mm aggregate	cum	0.36	487.00	175.32
					529.83
					100289.69
	Rate per m = (a+b+c+d+g)/48 =				2089.37
				or say	2089.37
	Sub Analysis for Rate of Concrete				
	(a) Material				
	Cement	t	0.40	3900.00	1560.00
	Coarse sand	cum	0.45	100.00	45.00
	20 mm aggregate	cum	0.54	1303.00	703.62
	10 mm aggregate	cum	0.36	894.00	321.84
					2630.46
	(b) Labour				
	Mate	day			
	Mason (1st Class)	day	0.12	475.00	57.00
	Mazdoor (Unskilled)	day	2.09	400.00	836.00
					893.00
	Add 40% area allowance on labour				357.20
					1250.20
	c) Machinery				
	Concrete mixer 0.4/0.28 cum capacity	hour	0.40	525.50	210.20
					4090.86
	Formwork @ 12%		12.00%		490.90
	Rate per cum				4581.76
					4582.00
14	RBR-SBST-9 page.No.565				
	Providing weep holes in brick masonry/stone masonry, plain/reinforced concrete abutment, wing wall, return wall with 100 mm dia AC pipe extending through the full width of the structures with slope of 1(V):20(H) towards drawing face complete as per drawing and technical specification Clauses 614, 709, 1204.3.7 MORD / 2706 & 2200 MORTH.				
	Unit = Nos				
	Taking output = 30 Nos				
	(a) Material				

S. No	Description	Unit	Quantity	Rate Rs.	Amount Rs.
1	2	3	4	5	6
	AC pipe 100 mm dia including wastage @ 5 per cent. Average length of weep hole is taken as one metre for the purpose of estimating (Page No-66 of SSR)	m	31.500	253.00	7969.50
	MS clamps	Nos.	30.000	195.00	5850.00
	Collar for AC pipe (average taking 10% of above pipe rate)		10.00%		796.95
	Cement mortar 1:3 (For rate refer to RBR-FNDN-5(i) page No.497)	cum	0.050	2057.00	102.85
					14719.30
	(b) Labour				
	Mate	day	-		
	Mason 1st Class	day	0.500	475.00	237.50
	Mazdoor (Unskilled)	day	0.280	400.00	112.00
					349.50
	Add 40% area allowance on labour				139.80
					489.30
	(a+b) =				15208.60
	c&d) Overheads & Contractors Profit (a+b) 13.615%				2070.65
	Cost for 30 Nos = (a+b+c+d)				17279.25
	Rate per No = (a+b+c+d)/30=				575.98
				or say	576.00
	CementMortar 1:3 (1 Cement: 3 sand)				
	Unit- 1 Cum				
	(a) Material				
	cement	T	0.48	3900.00	1872.00
	Sand	Cum	1.05	100.00	105.00
					1977.00
	(b) Labour				
	Mate	day			
	Mazdoor	day	0.20	400.00	80.00
	Add 40% area allowance on labour				
					80.00
	Total Material and labour (a+b)				2057.00
15	RBR-SPST-11 Page No.592				
	Drainage spouts complete as per drawing and technical specifications Clause 1209 MORD & 2705 MORTH				
	Unit = 1 No				
	(a) Material				
	Corrosion resistant structural steel grating including 5 per cent wastage (Page No-29 of SSR)	kg	4.00	50.14	200.55
	G I pipe 100 mm dia Page No-12of SSR) B-class	m	6.00	675	4050.00
	GI bolt 10 mm Dia (Page No-30 of SSR)	each	6.00	19	114.00
	Galvanised MS flat clamp (Page No-29 of SSR)	each	2.00	165	330.00
					4694.55
	(b) Labour				
	For fabrication				
	Mate	day	-		
	Blacksmith, Welder etc. (Skilled)	day	0.02	475.00	9.50
	Mazdoor (Unskilled)	day	0.04	400.00	16.00
					25.50
	Add 40% area allowance on labour				10.20
					35.70
	For fixing in position				
	Mate	day	-		
	Mason (1st Class)	day	0.01	475.00	4.75
	Mazdoor (Unskilled)	day	0.21	400.00	84.00
					88.75
	Add 40% area allowance on labour				35.50
					124.25
					4854.50
	Add @ 5 per cent of cost of material and labour (a+b) for electrodes, gas cutting, sealant, anti-corrosive bituminous paint, mild steel grating etc.		5.00%		242.73
	Total				5097.23
	c&d) Overheads & Contractors Profit at 13.615%				693.99
	Rate per 1 No. = a+b+c+d				5791.21

S. No 1	Description 2	Unit 3	Quantity 4	Rate Rs. 5 or say	Amount Rs. 6 5791.20
16	RBR-SPST-17(III)(B) Page.No.598 (B) Providing and fixing in position 20 mm thick premoulded joint filler in expansion joint for fixed ends of simply supported spans not exceeding 10 m to cater for a horizontal movement upto 20 mm, covered with sealant complete as per drawing and technical specifications MORTH				
	Unit = Running m				
	Taking output = 12 m				
	(a) Material				
	Premoulded joint filler 20 mm thick and 300 mm deep (BMT-U-02 page 71 Bldg SSR)	sqm	3.6	735.00	2646.00
	(b) Labour (Item 801 Page 68 of SSR Buildings)				
	Mate	day	-		
	Mazdoor (Unskilled)	day	0.21	400.00	84.00
	Mazdoor (Skilled)	day	0.1	400.00	40.00
					124.00
	Add 40% area allowance on labour				49.60
					173.60
	(a+b)				2819.60
	c&d) Overheads & Contractors Profit 13.615%				383.89
	Cost for 12 m = a+b+c+d				3203.49
	Rate per m = (a+b+c+d)/ 12 =				266.96
				or say	267.00
17	BMS-W-20 Mastic Pad 2' 0"x4' 0" (0.60M x 1.20 M) of 1" (25.40 MM) thick (Page No-75 of Bldg. SSR 201 1-12 item No.1102)	Sqm	1.00	839.00	839.00
18	RBR-PTWK-7 (ii) (ii) Gravel Backing for Revetments as per APSS				
	Unit = cum				
	(a) Material				
	Gravel	cum	1.16	110.00	127.60
	(b) Labour				
	Mazdoor (Unskilled)	day	1.1	400.00	440.00
	Add 40% area allowance on labour				176.00
					616.00
	(a+b)				743.60
	c&d) Overheads & Contractors Profit @ 13.615%				101.24
					844.84
	g)Add Conveyance charges				
	Gravel	cum	1.16	216.40	251.02
	Rate per cum				1096.87
					1096.00
19	A RBR-SBBS-2(A) P.318 Granular Sub-Base with Coarse Graded Material (Table:- 400- 2) Construction of granular sub-base by providing coarse graded material, spreading in uniform layers with motor grader on prepared surface, mixing by mix in place method with rotavator at OMC, and compacting with vibratory roller to achieve the desired density, complete as per clause 401 MORTH.				
	Unit = cum				
	Taking output = 300 cum				
	a) Labour				
	Mate	day	-		
	Mazdoor skilled	day	2.40	400.00	960.00
	Mazdoor	day	8.00	400.00	3200.00
					4160.00
	Add 40% area allowance on labour				1664.00
					5824.00
	b) Machinery				
	Mortar Grader 110 HP @ 50 cum per hour	hour	6.00	3020.40	18122.40
	Vibratory roller 8 -10 tonne (P-28)	hour	6.00	2736.20	16417.20

S. No	Description	Unit	Quantity	Rate Rs.	Amount Rs.
1	2	3	4	5	6
	Water tanker € KL capacity	hour	3.00	655.20	1965.60
					36505.20
	c) Material				
	For coarse graded Granular sub-base Materials per table 400-2				
	For grading-I Material (CBR Value minimum 30)				
	53 mm to 26.5 mm @ 35 per cent	cum	134.40	813.00	109267.20
	26.5 mm to 4.75 mm @ 45 per cent	cum	172.80	593.00	102470.40
	2.36 mm below @ 20 per cent (Coarse Sand)	cum	76.80	333.00	25574.40
	cost of water(urban)M189a	KL	18.00	103.00	1854.00
					239166.00
	(a+b+c)				281495.20
	d&e) Overheads & Contractors Profit @ 13.615%				38325.57
					319820.77
	g)Add Conveyance charges				
	53 mm to 26.5 mm @ 35 per cent	cum	134.40	487.00	65452.80
	26.5 mm to 4.75 mm @ 45 per cent	cum	172.80	487.00	84153.60
	2.36 mm below @ 20 per cent (Coarse Sand)	cum	76.80	487.00	37401.60
					187008.00
	Cost for 300 cum = a+b+c+d+e				506828.77
	Rate per cum = (a+b+c+d+e)/300=				1689.43
					1689.00
	 Assistant Executive Engineer Special Section Vijayawada			 Dy. Executive Engineer Special Sub Division Vijayawada	

AS PER IRRIGATION STANDARD DATA

Laying and jointing 1000 mm dia. NP- 3 class or IRC standard hume pipes in CM 1 : 2 proportion by volume including cost of all materials(excluding pipes and collars), machinery,labour, aligning, packing joints with hemp, finishing, curing etc., complete with initial lead upto 50 m and all lifts.(Cement content: 49.5 kg / joint, FA : 0.05cum/joint, Hemp Yarn : 0.377kg/joint) IRR-CCDW-6-7

RATE ANALYSIS				UNIT :	10.00	Joints
Sl No	Particulars	Unit	Quantity	Rate in Rs.	Amount in Rs.	
1	Cement	kg	495.00	3.90	1930.50	
2	Sand	cum	0.50	182.00	91.00	
3	Hemp yarn	kg	3.77	61.00	229.97	
Total cost of Materials				Rs:	2251.47	

B. MACHINERY:					
Sl No	Description	Unit	Quantity	Rate in Rs.	Amount in Rs.
1	Nil		0.00	0.00	0.00
	(Manual mixing)		0.00	0.00	0.00
Total hire charges of Machinery				Rs:	0.00

C. LABOUR:					
Sl No	Description	Unit	Quantity	Rate in Rs.	Amount in Rs.
1	Mason Cl- II	Day	3.00	440.00	1320.00
2	work inspector	Day	1.00	500.00	500.00
3	mazdoor	Day	6.00	400.00	2400.00
Total cost of Labour				Rs:	4220.00
Add 40% area allowance on labour					1688.00
Total cost of Labour				Rs:	5908.00
labour component/unit qty			422.00		
Add contractor's profit and overhead charges			0.13615	57.46	
labour component/unit qty (including contractor's profit)			479.46		

ABSTRACT:

A. Cost of Materials including seignorage charges	Rs:	2251.47
B. Hire charges of Machinery	Rs:	0.00
C. Cost of Labour	Rs:	5908.00
	Total Rs:	8159.47
D. Add for contractor's profit and overheads on (A+B+C)	Rs:	1110.91
Total cost for	10.00 Joints	Rs: 9270.38
E)Add conveyance charges		
Sand screened	cum	0.45
		203.40
		91.53
		9361.91
	(A+B+C+D)/10.0	Rs: 936.19
		936.19
		93.62
		1031.00

2 Providing and constructing 30 cm thick dry rubble stone pitching including cost of all materials, labour, hand packing, finishing etc., complete(rubble stones : 0.33 cum/sqm)

RATE ANALYSIS				UNIT:	100	Sqm.
IRR-CAW-8-4						
A. MATERIALS :						
Sl. No	Particulars	Unit	Quantity	Rate in Rs.	Amount in Rs.	
1	Uncoursed rubble stones at Quarry	Cum	33.00	303.00	9999.00	
Total Cost of Materials Rs.					9999.00	
B. MACHINERY :						
Sl. No	Particulars	Unit	Quantity	Rate in Rs.	Amount in Rs.	
1	Nil		0	0		
			0	0		
Total hire charges of Machinery				Rs.	0	
C. LABOUR :						
Sl. No	Particulars	Unit	Quantity	Rate in Rs.	Amount in Rs.	
1	Mason Class II	Day	6.60	440.00	2904.00	
2	Mazdoor	Day	29.70	400.00	11880.00	
Total Cost of Labour				Rs.	14784.00	
Add 40% area allowance on labour.					5913.60	
					20697.60	
labour component/unit qty			147.84			
Add contractor's profit and overhead charges			13.615%	20.13		

labour component/unit qty (including contractor's profit)

167.97

ABSTRACT :

A. Cost of Materials

Rs. 9999.00

B. Hire Charges of Machinery

Rs. 0

C. Cost of Labour

Rs. 20697.60

Total

Rs. 30696.60

D. Add for contractor's profit and overheads on (A+B+C)

13.615%

Rs. 4179.34

34875.94

E. Add Conveyance Charges:

1 Uncoursed rubble stones at Quarry

Cum

33.00

487.00

16071.00

Total cost for

100.00**Sqm.**Rs. **50946.94**

Rate per Sqm

(A+B+C+D+E)/100.0Rs. **509.47**say **509.00** *509.50*

Providing and laying insitu vibrated M-15 (28 days cube compressive strength not less than 15 N / sq mm) grade cement concrete using 40 mm down size approved, clean, hard graded aggregates for foundation filling including cost of all materials, machinery, labour, formwork, cleaning, batching, mixing, placing in position, levelling, vibrating, finishing, curing etc., complete with initial lead upto 50 m and all lifts. (Cement content: 260 kg / cum with use of super plasticiser (0.4% by wt. of cement), CA : 0.90cum, Blending Ratio of CA-50:30:20, FA : 0.40 cum) **for Levelling course**

IRR-CCDW-2-3

DATA:

RATE ANALYSIS

UNIT : 15.38 cum

A. MATERIALS:

SI No	Particulars	Unit	Quantity	Rate in Rs.	Amount in Rs.
1	Cement for mix	kg	3998.80	3.90	15595.32
	Cement for incidentals @ 3 kg / cum	kg	46.14	3.90	179.95
2	Coarse aggregate 40-20 mm	cum	6.92	1133.00	7840.36
	Coarse aggregate 20-10 mm	cum	4.15	1182.00	4905.30
	Coarse aggregate 10 mm below	cum	2.77	860.00	2382.20
3	Fine aggregate (Un-Screened)	cum	6.15	100.00	615.00
4	Super Plasticizer	kg	16.00	43.00	688.00
5	Use rate of shuttering for 40 uses	sqm	15.38	228.86	3519.87
	Scaffolding @ of shuttering	10%			351.99
6	Sundries	LS	0.50	31.00	15.50
Total cost of Materials				Rs:	36093.48

B. MACHINERY:

SI No	Description	Unit	Quantity	Rate in Rs.	Amount in Rs.
1	Concrete mixer 300/200 ltr (diesel)	Hour	8.00	43.50	348.00
	Fuel / Energy charges	Hour	8.00	104.20	833.60
2	5 hp pump (diesel)	Hour	0.50	8.30	4.15
	Fuel / Energy charges	Hour	0.50	104.20	52.10
3	Water tanker 8000 ltr	Hour	1.00	359.00	359.00
	Fuel / Energy charges	Hour	1.00	393.70	393.70
4	Needle vibrator 40 mm dia (petrol)	Hour	8.00	6.20	49.60
	Fuel / Energy charges	Hour	8.00	23.20	185.60
Total hire charges of Machinery				Rs:	2225.75

C. LABOUR:

SI No	Description	Unit	Quantity	Rate in Rs.	Amount in Rs.
1	Crew for Concrete mixer	Hour	8.00	241.80	1934.40
2	Crew for Pump	Hour	0.50	122.30	61.15
3	Crew for Water tanker	Hour	1.00	193.10	193.10
4	Crew for Vibrator	Hour	8.00	174.10	1392.80
5	Mason Class-I	Day	1.00	475.00	475.00
6	work inspector	Day	1.00	500.00	500.00
7	mazdoor				0.00
	for batching materials	Day	11.00	400.00	4400.00

	for loading mortar pans	Day	4.00	400.00	1600.00
	for laying	Day	3.00	400.00	1200.00
	for conveying concrete	Day	15.38	400.00	6152.00
8	for cleaning/ washing/ curing	Day	1.00	400.00	400.00
9	Labour cost for shuttering	sqm	15.38	99.60	1531.85
	Labour cost for scaffolding @	10%			153.18
	Total cost of Labour			Rs:	19993.48
	Add 40% area allowance on labour		40%		7997.39
	Total cost of Labour				27990.88
	labour component/unit qty		1299.97		
	Add contractor's profit and overhead charges	0.13615	9548.05		
	labour component/unit qty (including contractor's profit)		10848.01		
ABSTRACT:					
	A. Cost of Materials			Rs:	36093.48
	B. Hire charges of Machinery			Rs:	2225.75
	C. Cost of Labour			Rs:	27990.88
				Total Rs:	66310.11
	D. Add for contractor's profit and overheads on (A+B+C)		13.615%	Rs:	9028.121
	Total cost for	15.38 cum		Rs:	75338.23
	E. Add Conveyance Charges:				
	Coarse aggregate 40-20 mm	cum	6.92	487.00	3370.04
	Coarse aggregate 20-10 mm	cum	4.15	487.00	2021.05
	Coarse aggregate 10 mm below	cum	2.77	487.00	1348.99
	Fine aggregate (Un-Screened)	cum	6.15	203.40	1250.91
	cost for conveyance charges				7990.99
	Total cost for	16.36 cum		Rs:	83329.22
	Rate per Cum	(A+B+C+D)/16.36		Rs.	5418.02
				say	5418.00

Providing and laying insitu vibrated M-20 (28 days cube compressive strength not less than 20 N / sq mm) grade cement concrete using 20 mm down size approved, clean, hard, graded aggregates for **deck slab** including cost of all materials, machinery, labour, formwork, scaffolding, cleaning, batching, mixing, placing in position, levelling, vibrating, finishing, curing etc., complete with initial lead upto 50 m and all lifts. If water is to be brought from other place add only lead charges @ 500 ltr / cum. (Cement content: 330 kg / cum with use of super plasticiser (0.4% by wt. of cement), CA : 0.80 cum, Blending Ratio of CA:-65:35, FA : 0.45 cum)

4

				IRR-CCDW-2-24	
DATA:		RATE ANALYSIS		UNIT :	15.76 cum
A. MATERIALS:					
Sl No	Particulars	Unit	Quantity	Rate in Rs.	Amount in Rs.
1	Cement for mix	kg	5200.80	3.90	20283.12
	Cement for incidentals @ 5 kg / cum	kg	78.80	3.90	307.32
2	Coarse aggregate 20-10 mm	cum	8.20	1182.00	9686.73
	Coarse aggregate 10 mm below	cum	4.41	860.00	3795.01
3	Fine aggregate (Un-Screened)	cum	7.09	100.00	709.20
4	Super Plasticizer	kg	20.80	43.00	894.54
5	Use rate of shuttering for 30 uses	sqm	39.40	281.54	11092.68
	Scaffolding @ of shuttering	250%			27731.69
6	Sundries	LS	1.00	31.00	31.00
	Total cost of Materials			Rs:	74531.28
B. MACHINERY:					
Sl No	Description	Unit	Quantity	Rate in Rs.	Amount in Rs.
1	Concrete mixer 300/200 ltr (diesel)	Hour	8.00	43.50	348.00
	Fuel / Energy charges	Hour	8.00	104.20	833.60
2	5 hp pump (diesel)	Hour	0.50	8.30	4.15
	Fuel / Energy charges	Hour	0.50	104.20	52.10
3	Needle vibrator 40 mm dia (petrol)	Hour	8.00	6.20	49.60
	Fuel / Energy charges	Hour	8.00	23.20	185.60
	Total hire charges of Machinery			Rs:	1473.05
C. LABOUR:					
Sl No	Description	Unit	Quantity	Rate in Rs.	Amount in Rs.
1	Crew for Concrete mixer	Hour	8.00	241.80	1934.40
2	Crew for Pump	Hour	0.50	122.30	61.15
3	Crew for Needle vibrator	Hour	8.00	174.10	1392.80
4	Mason Class-I	Day	2.00	475.00	950.00
5	work inspector	Day	1.00	500.00	500.00
6	mazdoor				0.00
	for batching materials	Day	11.00	400.00	4400.00
	for loading mortar pans	Day	4.00	400.00	1600.00
	for laying	Day	4.00	400.00	1600.00
	for conveying concrete	Day	15.76	400.00	6304.00
	for cleaning/ washing/ curing	Day	1.00	400.00	400.00
7	Labour cost for shuttering	sqm	39.40	99.60	3924.24
	Labour cost for scaffolding @	250%			9810.60
	Total cost of Labour			Rs:	32877.19
	Add 40% area allowance on labour	40%			13150.88
	Total cost of Labour				46028.07
labour component/unit qty			2086.12		
Add contractor's profit and overhead charges			13.615%	284.02	
labour component/unit qty (including contractor's profit)			2370.14		
ABSTRACT:					
A. Cost of Materials				Rs:	74531.28
B. Hire charges of Machinery				Rs:	1473.05
C. Cost of Labour				Rs:	46028.07
				Total Rs:	122032.39
D. Add for contractor's profit and overheads on (A+B+C)			13.615%	Rs:	16614.71
Total cost for			15.76 cum	Rs:	138647.10
E. Add Conveyance Charges:					
	Coarse aggregate 20-10 mm	cum	8.20	487.00	3991.06
	Coarse aggregate 10 mm below	cum	4.41	487.00	2149.03
	Fine aggregate (Un-Screened)	cum	7.09	203.40	1442.51
					7582.61
Total cost for			15.76	Rs:	146229.71
Rate per Cum			(A+B+C+D)/15.76	Rs:	9278.54
				say	9278.50

- 5 Providing, fabricating and **placing in position reinforcement steel bars for RCC works** including cleaning, straightening, cutting, bending, hooking, lapping, welding wherever required, tying with 1.25 mm dia soft annealed steel wire, including cost of all materials, machinery, labour etc., complete with initial lead upto 50 and all lifts.

DATA:

IRR-CCDW-2-1

Lap jointing considered assuming all reinforcement below 36 mm dia.

Quantity of binding wire per tonne :

8 kg

Wastage of steel :

5 percent

Cleaning, straightening :

1 Bar benders & 2 mazdoors

Marking, cutting, bending & stacking :

2 Bar benders & 4 mazdoors

Fixing & tying with binding wire :

2 Bar benders & 4 mazdoors

Checking / correcting & misc. :

1 Bar benders & 1 mazdoors

DATA: RATE ANALYSIS

UNIT: 1000.00 kg

A. MATERIALS:

Sl No	Particulars	Unit	Quantity	Rate in Rs.	Amount in Rs.
1	Rein.Steel with 5 % wastage	kg	1050.00	43.50	45675.00
2	Binding wire 1.25 mm dia	kg	8.00	53.00	424.00
3	Sundries (chairs / spacers etc)	LS	3.00	31.00	93.00
	Total cost of Materials			Rs:	46192.00

B. MACHINERY:

Sl No	Description	Unit	Quantity	Rate in Rs.	Amount in Rs.
1	Nil		0.00	0.00	0.00
			0.00	0.00	0.00
	Total hire charges of Machinery			Rs:	0.00

C. LABOUR:

Sl No	Description	Unit	Quantity	Rate in Rs.	Amount in Rs.
1	work inspector	Day	1.00	500.00	500.00
2	Bar bender	Day	6.00	600.00	3600.00
3	mazdoor	Day	11.00	400.00	4400.00
	Total cost of Labour			Rs:	8500.00
	Add 40% area allowance on labour	40%			3400.00
	Total cost of Labour				11900.00

labour component/unit qty

8.50

Add contractor's profit and overhead charges

0.13615 1.16

labour component/unit qty (including contractor's profit)

9.66

ABSTRACT:

A. Cost of Materials Rs: 46192.00

B. Hire charges of Machinery Rs: 0.00

C. Cost of Labour Rs: 11900.00

Total Rs: 58092.00

D. Add for contractor's profit and overheads on (A+B+

13.615% Rs: 7909.2258

Total cost for

1000.00 kg Rs: 66001.23

Rate per kg

(A+B+C+D)/1000.0 Rs: 66.001

say 66.000

A. I. C. E.
21/2/19
Assistant Executive Engineer
Special Section
Vijayawada

Dy. Executive Engineer
Special Sub Division
Vijayawada

Executive Engineer
Water Resources Department
Special Division, Vijayawada.

DATA FOR BRIDGE APPROACH PORTION						
Name of Work:Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta Drainage System .						
Sl. No	Quantity	Description of Item	Rate	per	Amount	
1	Providing, Laying, Spreading and compacting graded HBG crushed stone aggregate to Wet Mix macadem specification including cost,seigniorage of all materials and including premixing the material with water at OMC in Mechanical mix plant carriage of mixed material by tipper to site , laying in uniform layers with paver in base courses on well prepared surface and compacting with Vibratory roller to acheive the desired density etc., as directed by the Engineer-in-Charge and as per MoRT&H specification.406 (4 th revision) for finished item of work.					
	Unit = cum		Page 110,111 of MoRT&H SDB			RBR-SBBS-11
	Taking output = 225 cum					
	(A) Labour					
	nos.	Mate		day	0.00	
	2.48	nos.	Mazdoor skilled	400.00	day	992.00
	10.00	nos.	Mazdoor	400.00	day	4000.00
		Total			4992.00	
		Add 40% area allowance on labour (Page No-7 of SSR)				1996.80
		Total			6988.80	
	(B) Machinery					
	9.00	hr	Wet mix plant of 75 tonne hr. capacity	1679.70	hr	15117.30
	6.00	hr	Electric generating set 125 KVA	1543.10	hr	9258.60
	6.00	hr	front end loader 1 cum capacity	1594.70	hr	9568.20
	6.00	hr	Mechanical Paver finisher 100 TPH	2290.60	hr	13743.60
	3.90	hr	Vibratory roller 8-10 T	2736.20	hr	10671.18
		Total			58358.88	
	(C) Material					
	89.10	cum	45 to 22.40mm IRC&MoRT&H HBG M/C metal@ 30% (Av. Of 40-45, 25-27, 19-22mm)	798.73	cum	71167.14
	118.80	cum	22.4 to 2.36mm IRC&MoRT&H HBG M/C metal@40% (Av. Of 19-22mm, 12-14mm, 9.5-11.2mm, 5-7mm, 2.36-5mm	615.62	cum	73135.66
	89.10	cum	2.36mm to 75 micron @ 30% (2.36mm & below)	293.35	cum	26137.49
		Total			170440.28	
		Total (A)+(B)+(C)				235787.96
		Add 13.615% towards Contractor's Profit on Rs				32102.53
		Total			267890.49	
	D)Add conveyance charges					
	89.10	cum	45 to 22.40mm IRC&MoRT&H HBG M/C metal@ 30%	487.00	cum	43391.70
	118.80	cum	22.4 to 2.36mm IRC&MoRT&H HBG M/C metal@40%	487.00	cum	57855.60
	89.10	cum	2.36mm to 75 micron @ 30% (2.36mm & below)	487.00	cum	43391.70
					144639.00	
		Total for 225 cum				412529.49
		Rate per 1 cum				1833.46
		Rate per 1 cum				1833.00
						Cum
2	Providing and laying surface dressing in single coat using 6 mm nominal size IRC HBG machine crushed stone aggregates @ 0.004 cum / sqm on a layer of bitumen binder of 80/100 grade @ 0.75 kg / sqm including cost, seigniorage and conveyance of all materials to work site and laid on prepared surface and rolling with 8-10 T Power Road Roller etc., complete for finished item of work as per MoRT&H Specification 510 (4 th Revision) and as directed by the Engineer-in-charge.					
	Unit = sqm		Page 141,142 of MoRT&H SDB			RBR-BASC-11-II
	Taking output = 9000 sqm					

Sl. No	Quantity	Description of Item	Rate	per	Amount
	(A) Labour				
	nos.	Mate		day	0.00
9.00	nos.	Mazdoor unskilled	400.00	day	3600.00
2.44	nos.	Mazdoor skilled	400.00	day	976.00
		Total			4576.00
		Add 40% area allowance on labour			1830.40
		Total			6406.40
	(B) Machinery				
7.20	hr	Mechanical broom @ 1250 sqm/hr.	381.80	hr	2748.96
7.20	hr	Air compressor 250 cfm(7 cmm)	671.40	hr	4834.08
6.00	hr	Hydraulic self propelled chip spreader @ 1500 sqm/hr.	2743.70	hr	16462.20
6.00	hr	Tipper 10Tonne capacity for carriage of stone chips from stock pile on road side to chip spreader. (Rate of Tipper 5 cum	950.30	hr	5701.80
6.00	hr	Front end loader 1 cum bucket capacity	1594.70	hr	9568.20
6.00	hr	Bitumen pressure distributor @1750 sqm/hr	1527.10	hr	9162.60
6.00	hr	smooth wheeled roller 8 Ton	1329.90	h	7979.40
		Total			56457.24
	(C) Material				
6.75	MT	Bitumen 80/100 @ 0.75 Kg / sqm	34260.60	MT	231259.05
36.00	cum	Crushed stone chippings 6 mm nominal size @0.004 cum/sqm	480.00	cum	17280.00
		Total			248539.05
		Total (A)+(B)+ (C)			311402.69
		Add 13.615% towards Contractor's Profit on Rs	305018.06		42397.48
		Total			353800.17
	D)Add conveyance charges				
36.00	cum	Crushed stone chippings 6 mm nominal size @0.004 cum/sqm	487.00	cum	17532.00
		Total per 9000 Sq.M			371332.17
		Total cost per SqM			41.26
		Rate to be adopted.			41.00
					1 Sqm
3	Providing and applying tack coat with bitumen Emulsion (Medium setting) (Bulk) using Emulsion pressure distributor at the rate of 0.20 kgs per sqm on the prepared bituminous/granular surface cleaned with mechanical broom for finished item of work as per MoRT&H Specification 503 (4 th revision) and as directed by the Engineer-in-Charge.				
	unit = sqm		Page 124 of MoRT&H SDB		RBR-BASC-2-V
	Taking output = 3500 sqm				
	(A) Labour				
	nos	Mate		day	0.00
2.08	nos	Mazdoor un skilled	400.00	day	832.00
		Total			832.00
		Add 40% area allowance on labour			332.80
		Total			1164.80
	(B) Machinery				
2.80	hr	Mechanical broom Hydraulic @ 1250	318.80	hr	892.64
2.80	hr	Air compressor 250 cfm	671.40	hr	1879.92
2.00	hr	Emulsion pressure distributor @1750	875.60	hr	1751.20
		Total			4523.76
	(C) Material				
0.70	MT	Bitumen Emulsion @ 0.20 Kgs/sqm	46686.00	MT	32680.20
		Total			32680.20
		Total (A)+(B)+ (C)			38368.76
		Add 13.615% towards Contractor's Profit on Rs	38368.76		5223.91
		Total			43592.67
		Total per 3500 Sq.M			43592.67
		Total cost per SqM			12.46
		Rate to be adopted			12.00
					1 Sqm

Sl. No	Quantity	Description of Item	Rate	per	Amount
4	Providing and laying of 50mm thick Bituminous Macadam with hot mix plant producing an average output of 37.5 tonnes per hour using hard blasted granite crushed aggregates of Grading - II as per table 500-4 of MoRT&H specification 504 (4 th Revision) premixed with bituminous binder of 60/70 grade @ 3.3% by weight of total mixture, transported to site, laid over a previously prepared surface with mechanical paver finisher to the required grade, level and alignment and rolled as per Clauses 501.6 and 501.7 to achieve the desired compaction for finished item of work as directed by the Engineer-in-Charge including hire and operational charges all T&P and all other contingent charges necessary including cost of seigniorage charges on all materials etc., complete and as per MoRTH specification No. 504(4 th Revision) RBR-BASC-3.				
	Quantity	Description of Item	Rate	per	Amount
	Unit = cum				
	Taking out put = 205 cum (450 tonne) Page 125,126,127 of MoRT&H SDB				
	(A)	Labour			
		nos.	Mate		
				420.00	nos.
					0.00
	14.00	nos.	Mazdoor working with HMP, mechanical broom, paver, roller, asphalt cutter and assistance for setting out lines, levels and layout of construction.	400.00	nos.
					5600.00
	5.84	nos.	Mazdoor skilled	400.00	nos.
					2336.00
			Total		7936.00
			Add 40% area allowance on labour		3174.40
			Total		11110.40
	(B)	Machinery			
	12.00	hr	Hot Mix Plant 40 to 60 TPH	19546.00	hr
					234552.00
	2.20	hr	Air Compressor 250 cfm	671.40	hr
					1477.08
	6.00	hr	Mechanical Paver finisher 100 TPH	2290.60	hr
					13743.60
	6.00	hr	Generator 250 KVA	2162.60	hr
					12975.60
	6.00	hr	Front end loader 1 cum bucket capacity	1594.70	hr
					9568.20
	3.90	hr	Smooth wheeled roller 8-10 tonnes for	1329.90	hr
					5186.61
	3.90	hr	Vibratory roller 8 tonnes for intermediate	2736.20	hr
					10671.18
	3.90	hr	Finish rolling with 6-8 tonnes smooth wheeled tandem roller	1856.50	hr
					7240.35
			Total		295414.62
	(C)	Material			
	14.85	MT	Bitumen 60/70 @ 3.3 % of weight of mix=205x2.2=450 tonne	35204.60	MT
					522788.31
			ii) Aggregate		
			Total weight of mix = 450 tonnes		
			weight of bitumen - 14.85 tonnes		
			Weight of aggregate = 450-14.85 =		
			Taking density of aggregate = 1.5		
			Volume of aggregate = 290.1 cum		
			Grading - II (19 mm nominal size)		
			116.04 Cum of 25-10mm @ 40%		
	116.04	cum	IRC & MoRT&H HBG M/C chips 25 to 10mm @ 40%	800.95	cum
					92942.24
			Average rate of IRC&MoRT&H HBG M/C metal of sizes 25-27mm, 19-22mm, 12-14mm, 9.5-11.2mm		
	116.04	cum	IRC & MoRT&H HBG M/C chips 10 to 5mm @ 40%	552.38	cum
					64097.60
			Average rate of IRC&MoRT&H HBG M/C metal of sizes 9.5-11.2mm, 5 to 7mm		
	58.02	cum	IRC & MoRT&H HBG M/C chips 5mm and below @ 20%	313.18	cum
					18170.41

Sl. No	Quantity		Description of Item	Rate	per	Amount
			Average rate of IRC&MoRT&H HBG M/C metal of sizes 2.36 to 5mm and 2.36mm& below			
			Total			697998.56
			Total (A)+(B)+ (C)			1004523.58
			Add 13.615% towards Contractor's Profit on Rs	1004523.58		136765.88
			Total			1141289.46
			D)Add conveyance charges			
	116.04	cum	IRC &MORT&H HBG M/C chips 25 to 10mm @ 40%	487.00	cum	56511.48
	116.04	cum	IRC &MORT&H HBG M/C chips 10 to 5mm @ 40%	487.00	cum	56511.48
	58.02	cum	IRC &MORT&H HBG M/C chips 5mm and below @ 20%	487.00	cum	28255.74
			Total			141278.70
			Total per 205 Cum			1282568.16
			Total cost per Cum			6256.43
			Rate to be adopted			6256.00
						1 Cum
5	Providing 25 mm thick compacted Semi-Dense Bituminous Concrete by hot mix plant using hard blasted granite crushed aggregates of Grading -2 as per table 500-15 of specification 508 of MoRT&H (4 th Revision) premixed with bituminous binder 60/70 grade @ 5% of mix and filler, transporting the hot mix to work site, laying with mechanical paver finisher to the required grade, level and alignment, rolling with smooth wheeled,vibratory and tandem rollers to achieve the desired compaction as per MoRT&H Specification 508 (4 th Revision) complete for finished item of work in all respects as directed by Engineer-in-charge.					
Unit = cum			Page 135,136,137 of MoRT&H SDB			RBR-BASC-8
Taking output = 195 cum (450 tonnes.)						
(A) Labour						
	nos.	Mate			day	0.00
14.00	nos.	Mazdoor working with HMP, Mechanical broom, paver,roller,asphalt cutter and assistance for setting outlines,levels and	400.00		day	5600.00
5.84	nos.	Skilled Mazdoor for checking line and levels.	400.00		day	2336.00
		Total				7936.00
		Add 40% area allowance on labour				3174.40
		Total				11110.40
(B) Machinery						
11.00	hr	Hot mix plant 40 to 60 TPH	19546.00		hr	215006.00
6.00	hr	Mechanical Paver finisher @ 100 TPH	2290.60		hr	13743.60
6.00	hr	Generator 250 KVA	2162.60		hr	12975.60
6.00	hr	Front end loader 1 cum capacity	1594.70		hr	9568.20
3.90	hr	Power road roller 8 tonne for initial break down	1329.90		hr	5186.61
3.90	hr	Vibratory roler 8 tonne for intermediate	2736.20		hr	10671.18
3.90	hr	Finish rolling with 6-8 tonne smooth wheeled tandem roller.	1856.50		hr	7240.35
		Total				274391.54
(C) Material						
22.50	M.T	(i) Bitumen @ 5% by weight of mix	35204.80		MT	792103.50
		(ii) Aggregate For Grading II				
162.450	cum	IRC &MORT&H HBG M/C chips 9.5 tp 4.75mm @ 57%	466.03		cum	75707.12
		Average rate of IRC&MoRT&H HBG M/C metal of sizes 9.5-11.2, 5-7mm,2.36 - 5mm				
116.850	cum	IRC &MORT&H HBG M/C chips 4.75 mm and below @ 41%	313.18		cum	36594.50
		Average rate of IRC&MoRT&H HBG M/C metal of sizes 2.36 - 5mm and 2.36mm& below			cum	
5.70	cum	Filler @ 2% of weight of aggregates.	860.00		cum	4902.00
		Total				909307.11
		Total (A)+(B)+ (C)				1194809.05
		Add 13.615% towards Contractor's Profit on Rs	1194809.05			162673.25
		Total				1357482.31

Sl. No	Quantity		Description of Item	Rate	per	Amount
	D)Add conveyance charges					
	162.450	cum	IRC &MORT&H HBG M/C chips 9.5 to 4.75mm @ 57%	487.00	cum	79113.15
	116.850	cum	IRC &MORT&H HBG M/C chips 4.75 mm and below @ 41%	487.00	cum	56905.95
	5.70	cum	Filler @ 2% of weight of aggregates.	487.00	cum	2775.90
						138795.00
			Total per 195 Cum			1496277.31
			Rate per 1 cum			7673.22
			Rate to be adopted			7673.00
						1 Cum
6	Providing and laying of filter media using 50% of 150 MM IRC soling stone and 50% of 40 mm HBG metal satisfying the requirements laid down in clause 2504.2.2 of MoRT&H specifications to a thickness of not less than 600 mm with smaller size towards the soil and bigger size towards the wall and provided over the entire surface behind the abutment, wing wall and return wall to the full height compacted to a firm condition including cost and conveyance of all metal seignorage charges and all labour charges as directed by the departmental officers as per drawing and technical specifications as per clause 710.1.1 of IRC.78 and Clause 2200 of MoRTH&H (4th Revision) for finished item of work.					
	Unit = cum		Page 463 of MoRT&H SDB			
	Taking output = 10 cum					
	(A) Labour					
	nos.	Mate			day	0.00
	7.00	nos.	Mazdoor for filling watering, ramming etc	400.00	day	2800.00
	1.32	nos.	Mason	400.00	day	528.00
			Total			3328.00
			Add 40% area allowance on labour			1331.20
			Total			4659.20
	(B) Machinery					
	0.06	Hr	Water Tanker 6KL capacity	655.20	hr	39.31
	(C) Material					
			Filter media of stone aggregate conforming to clause 2504.2.2 of MoRT&H			
	6.00	Cum	Cost of 150MM size IRC soling stone (HBG)	90.50	Cum	543.00
	6.00	Cum	Cost of 40mm size IRC HBG metal	808.00	Cum	4848.00
			Total			5391.00
			Total (A)+(B)+ (C)			10089.51
			Add 13.615% towards Contractor's Profit on Rs			1373.69
			Total			11463.20
	D)Add conveyance charges					
	6.00	Cum	Cost of 150MM size IRC soling stone (HBG)	487.00	Cum	2922.00
	6.00	Cum	Cost of 40mm size IRC HBG metal	487.00	Cum	2922.00
			Total			5844.00
			Total per 10 Cum			17307.20
			Rate per 1 cum			1730.72
			Rate per 1 cum			1731.00
						1 Cum
			BLD-CSTN-10-4			
7	Painting to new walls with 2 coats of water proof cement paint (Snowcem)of approved brand and shade over a base coat of approved cement primer grade I making making 3 coats in all to give an even shade after thoroughly brushing the surface to remove all dirt and remains of loose powdered materials, including cost and conveyance of all materials to work site and all operational, incidental, labour charges etc. complete for finished item of work as per SS 912 for internal walls					
	a) labour					
	for cement primer					
	0.21	day	1st class painter	550.00	day	115.50
	0.49	day	2nd class painter	440.00	day	215.60
	for water proof cement paint					
	0.15	day	1st class painter	550.00	day	82.50
	0.36	day	2nd class painter	440.00	day	154.00
	1.50	day	Mazdoor (unskilled)	400.00	day	600.00
			Total			1167.60
			Add 40% area allowance on labour			467.04
			Total			1634.64
	b) Machinery-- NIL					
	c) Material					
	1.00	kg	Cost of Cement Primer	141.00	kg	141.00

[illegible]

COST OF BITUMEN / MT

Name of work: Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta Drainage System .

Ref :-		For Emulsion (Bulk) :- IOCL Price List With Effect From Dt. 1-02-2019					
		For Bulk Bitumen 80/100, 60/70:- IOCL Price List With Effect From Dt. 1-02-2019					
				Bulk 80/100 Grade (VG 10)	Bulk 60/70 Grade (VG 30)	VG 40	Emulsion (Bulk)
				Chennai	Chennai		Chennai
Basic Price of Chennai				27720.00	28520.00	29190.00	38250.00
GST @18%				4989.60	5133.60	5254.20	6885.00
				32709.60	33653.60	34444.20	45135.00
Lead from Chennai		470.000	Kms				
Lead charges for Bulk							
470.000	2	940.000					
			1.65	1551.00	1551.00	1551.00	1551.00
			per KM				
			TOTAL	34260.60	35204.60	35995.20	46686.00
 Asst Executive Engineer Special Section Vijayawada		 Dy Executive Engineer Special Sub Division Vijayawada		 Executive Engineer Water Resources Department Special Division, Vijayawada.			

ΕΒΕΤΗ: ΔΙΑΚΡΙΣΗ: ΔΙΠΛΩΜΑΤΑ:
ΜΕΤΕΡ: ΚΑΝΟΝΙΣΤΕΣ ΡΕΒΕΤΙΣΜΟΥ:
ΕΚΘΕΣΗ: ΕΥΡΩΠΑ

Name of work: Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna under Modernization of Krishna Delta Drainage System .

LEAD STATEMENT

S.No.	Description of material	Source of Supply Initial cost	Unit	Initial cost	SSR Page Reference	Blasting Charges	Machine crushing charges	Lead			Conveyance charges	Deduct Contractor Profit
								M.R	C.T.	Total		
1	Cement	Local	MT	3900				(G.O.Rt.No. 591 T,R& B (Roads-I) Dept Dated 25-10-2016)				
2	Steel	Local	MT	43500								
3	Gravel /Quarry spall(M-008)	Shabad	Cum	110	P.33			13	2	15	216.40	29.46
5	Sand for filling	Krishna	Cum	100	P.20			10	4	14	203.40	27.69
6	Sand for Mortar	Krishna	Cum	100	P.20			10	4	14	203.40	27.69
7	Sand (Sreened)	Krishna	Cum	182	P.20			10	4	14	203.40	27.69
8	Coarse Aggregate 10 - 4.75 MM	Donabanda	Cum	860	P.19 R&B			33	4	37	487.00	66.31
9	Coarse aggregate 20-10MM	Donabanda	Cum	1182	P.19 R&B			33	4	37	487.00	66.31
10	Coarse Aggregate 40-20 MM	Donabanda	Cum	1133	P.19 R&B			33	4	37	487.00	66.31
11	Coarse Aggregate 80-40 MM	Donabanda	Cum	622	P.19 R&B			33	4	37	487.00	66.31
12	HBG Metal 40MM(M-055)	Donabanda	Cum	808	P.28			33	4	37	487.00	66.31
13	HBG Metal 20 MM(M-053)	Donabanda	Cum	1303	P.27			33	4	37	487.00	66.31
14	HBG Metal 10 MM(M-051)	Donabanda	Cum	894	P.27			33	4	37	487.00	66.31
15	Coarse Graded Granular sub base material 53-26.5 MM(M-029)	Donabanda	Cum	813	P.26			33	4	37	487.00	66.31
16	Coarse Graded Granular sub base material 26.5 - 4.75 MM(M-026)	Donabanda	Cum	593	P.26			33	4	37	487.00	66.31
17	Rough Stone 225 MM thick	Donabanda	Cum	138	P.35			33	4	37	487.00	66.31
18	40 to 45 mm IRC and MoRTH HBG M/C metal	Donabanda	Cum	560	P.18 R&B	70	140	33	4	37	487.00	66.31
19	25 to 27 mm IRC and MoRTH HBG M/C metal	Donabanda	Cum	899.05	P.18 R&B	70	224.763	33	4	37	487.00	66.31
20	19 to 22 mm IRC and MoRTH HBG M/C metal	Donabanda	Cum	937.15	P.18 R&B	70	234.288	33	4	37	487.00	66.31
21	12 to 14 mm IRC and MoRTH HBG M/C metal	Donabanda	Cum	742.85	P.18 R&B	70	185.713	33	4	37	487.00	66.31

S.No.	Description of material	Source of Supply Initial cost	Unit	Initial cost	SSR Page Reference	Blasting Charges	Machine crushing charges	Lead			Conveyance charges	Deduct Contractor Profit
								M.R	C.T.	Total		
22	9.5 to 11.20 mm IRC and MoRTH HBG M/C metal	Donabanda	Cum	624.75	P.18 R&B	70	156.188	33	4	37	487.00	66.31
23	5 to 7 mm IRC and MoRTH HBG M/C metal	Donabanda	Cum	480	P.18 R&B	70	120	33	4	37	487.00	66.31
24	2.36 to 5 mm IRC and MoRTH HBG M/C metal	Donabanda	Cum	293.35	P.18 R&B	70	73.3375	33	4	37	487.00	66.31
25	Soling stone 150mm size of granite	Donabanda	Cum	90.5	P.19 R&B	70	22.625	33	4	37	487.00	66.31
26	HBG Stone chips 2.36mm and below	Donabanda	Cum	333	P.20 R&B			33	4	37	487.00	66.31
27	RCC pipes 900 MM dia NP3 class	Vijayawada	RM	4201	P.23 PH			10	0	10	255.19	4456.19
28	RCC collars 900 MM dia NP3	Vijayawada	each	759	P.24 PH			10		10	255.19	1014.19
29	RCC pipes 1000 MM dia NP3	Vijayawada	RM	5131	P.25 PH			10	0	10	295.76	5426.76
30	RCC collars 1000 MM dia NP3			947				10		10	295.76	1242.76
31	water charges urban(M189A)	Vijayawada	kilolitre	103								
32	300mm thick pitching stone	Donabanda		303				33	4	37	487.00	66.31

Certificates

Certified that the leads mentioned in the table are correct to the best of my

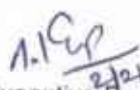
- 1 Knowledge.
- 2 Certified that the materials are available at the source of supply.
- 3 Certified that the seionerage charges proposed are actually payable.
- 4 Certified that the metal and stone are obtain by blasting.


Asst Executive Engineer
Special Section
Vijayawada


Dy. Executive Engineer
Special Sub Division
Vijayawada


Executive Engineer
Special Division
Vijayawada

STATEMENT SHOWING QUANTITY OF STEEL APPROCH ROAD PORTION									
II	Kerb& Parapet		Length		No of bars	Total Length		Weight	
	Kerb								
	Longitudinal 10MM		260		7	1820		1122.94	
	Rings 10MM dia		1.95		4 x 65/0.3	1690		1047.8	
	Parapet wall								
	Longitudinal 10MM								
			260		6	1560		967.2	
	Rings 10MM dia								
			1.95		4 x 65/0.3	1690		1047.8	
							Total	4185.74	
					Total Steel				4.19
								or say	4.19
									MT


 Asst Executive Engineer
 Special Section
 Vijayawada


 Dy. Executive Engineer
 Special Sub Division
 Vijayawada


 Executive Engineer
 Special Division
 Vijayawada

DETAILED ESTIMATE									
Name of work: Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta Drainage System.									
S.No	Description	Measurements						Contents of Area	
		No			L	B	D		
					Nos	Length of the Bar	Unit weight	Quantity in Kgs	
1	HYSD steel for Foundation								
	Foundation Cutoff, 12mm dia 200 c/c vertical (all round foundation) (56920+9620)/200=334 nos, Length of the Bar (1720+2220+420)=4.360mt	1	x	2	334.000	4.360	0.890	2592.11	
	12mm dia 200 c/c distribution (1720+420+1720)/200=20nos, Length of the Bar (56920+9620+56920+9620)=133.00mt	1	x	1	20.000	133.000	0.890	2367.40	
	b) for Raft								
	32mm dia 135c/c (bottom and top)								
	9700-80+9620/135=71.35+1=72 Nos, Length of the Bar (56920+1420+1420)=59.760mt	1	x	2	72.000	59.760	6.320	54386.38	
	32mm dia 67.5 c/c (EXT) (Bottom) 9620/67.5=142.52+1=144 Nos, Length of the Bar (9551+1420)=10.970mt	1	x	2	144.000	10.970	6.320	19967.16	
	16mm dia 180 c/c (Top & Bottom) distribution 56920/180=316.22+1=317 Nos, Length of the Bar (9620+1420+1420)=12.460mt	1	x	2	317.000	12.460	1.580	12481.43	
	16mm dia 200 c/c side distribution 1420/200=7+1=8Nos, Length of the Bar 9.620mt	1	x	2	8.000	9.620	1.580	243.19	
	c) For Wing								
	cut off vertical 12mm dia 200c/c 3420/200=18 Nos, Length of the Bar (1820+420+1820)=4.06mt	2	x	4	18.000	4.060	0.890	520.33	
	cut off distribution 12mm dia 200c/c 3260/200=17 Nos, Length of the Bar 4.22mt	2	x	2	17.000	4.220	0.890	255.39	
	for foundation main (top and bottom) 20mm dia 150c/c 3420/150=24 Nos, Length of the Bar (9620+720+720)=11.06mt	2	x	2	24.000	11.060	2.470	2622.55	
	for foundation distribution (top and bottom) 16mm dia 180c/c 9620/180=54 Nos, Length of the Bar (3420+720+720)=4.86mt	2	x	2	54.000	4.860	1.580	1658.62	
2	For Piers 25mm dia 64 nos, Length of bar (7422+1120+1120)=9.662 mt	1	x	4	64.000	9.662	3.860	9547.60	
	12mm 180c/c, 7422/180=42 nos Length of bar 18.70 mt	1	x	4	42.000	18.700	0.890	2796.02	
	Pier cap 16mm dia 6 nos each top and bottom, Length of bar (8620+220+220)=9.06 mt	1	x	4	12.000	9.060	1.580	687.11	
	stirrups 12mm dia 100c/c, 8620/100=87 nos	1	x	4	87.000	3.840	0.890	1189.32	
	For Abutment 25mm dia 175 c/c, 8370/175=49 nos, Length of bar (7422+1020+1020)=9.462 mt	2	x	2	49.000	9.462	3.860	7158.57	
	distribution 16mm dia 175c/c, 7422/175=43 nos, Length of the bar 11.10	2	x	2	43.000	11.100	1.580	3016.54	
	Abutment cap 16mm dia 6 nos each top and bottom, Length of bar (8370+220+220)=8.810 mt	1	x	2	12.000	8.810	1.580	334.08	
	stirrups 12mm dia 100c/c, 8370/100=85 nos, Length of the bar (1020+1020+220+220)=2.72mt	1	x	2	85.000	2.720	0.890	411.54	
	Dirt wall 16mm dia 12 nos both side, Length of bar (8370+420+420)=9.210 mt	1	x	2	12.000	9.210	1.580	349.24	
	stirrups 12mm dia 100c/c, 8370/100=85 nos, Length of the bar (1010+1010+420+420)=3.10mt	1	x	2	85.000	3.100	0.890	469.03	
	Parapet wall stirrups 12mm dia 250c/c, 57950/250=233 nos, Length of the bar (975+425+225+750+175)=2.55mt	1	x	2	233.000	2.550	0.890	1057.59	
	distribution 10mm dia 250c/c, 975/250=5 nos, Length of the bar 57.95	2	x	2	5.000	57.950	0.617	715.10	

S.No	Description	Measurements						Contents of Area	
		No			L	B	D		
	Approach slab : 12mm dia 150 C/c Top and Bottom, 8400/150=57 nos both side, Length of bar (3420+220+220)=3.860 mt	2	x	2	57.000	3.860	0.890	783.27	
	Approach slab distribution 12mm dia 150 C/c Top and Bottom, 3450/150=24 nos both side, Length of bar (3420+220+220)=3.860 mt	2	x	2	24.000	3.860	0.890	755.20	
3	for deck slab at 4067 KG/span (As per drawing MOST)	1	X	5	4067			20335.00	
4	Kerb and parapet								
								146699.36	
						add		110.14	
					Total Quantity of HYSD Steel			146810.00	Kgs

1.158
Assistant Executive Engineer
Special Section, Vijayawada

Deputy Executive Engineer,
Special Sub Division,

Executive Engineer
Special Division
Vijayawada

Name of work: Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta Drainage System .

Seigniorage Charges

Sl. No.	Item No.	Item	Quantity	Factor	Quantity of Materials (cum)					
					Metal		Sand		Earth/Gravel	
		Description of Item			factor	Quantity	factor	Quantity	factor	Quantity
1	2	3	4	5	6	7	8	9	10	11
2	3	Sand	496.00			0	1.05	520.8		
3	5	CC M10 40MSA	50.00		0.9	45	0.4	20		
4	6	RCC M30 20MSA	844.00		0.9	759.6	0.45	379.8		
5	7	RCC M20 20MSA	132.00		0.8	105.6	0.45	59.4		
6	8	RCC M25 20MSA	47.32		0.9	42.588	0.4	18.928		
7	9	RCC M25 20MSA(Sub)	385.00		0.9	346.5	0.4	154		
8	10	RCC M25 20MSA	93.00		0.9	83.7	0.4	37.2		
9	11	CC M15 20MSA			0.8	0	0.45	0		
10	13	300mm thick bed pitching	970.00		0.33	320.1		0		
11	15	RCC M30 20MSA	107.00		0.9	96.3	0.45	48.15		
12	16	wearing coat M30	45.00		0.9	40.5	0.45	20.25		
13	17	RCC M20 20MSA(deck slab)	345.50		0.8	276.4	0.45	155.475		
14	18	RCC M25 20MSA(railing)	123.60		0.9	111.24	0.4	49.44		
15	23	Gravel	1499.00						1.2	1798.8
16	24 & 28	GSB	787.50		1.28	1008		0		
17	29	WMM	255.00		1.32	336.6		0		
18	30	Surface dressing	1275.00		0.015	19.125		0		
19	32	Bituminous Macadam	64.00		1.415	90.56		0		
20	33	semi-Dense Bituminous Concrete	32.00		1.46	46.72		0		
		Total Quantity				3728.533		1463.443		1798.8
		Seigniorage charges per cum				75.00		50.00		30.00
		Amount of seigniorage charges in Rs.				279639.98		73172.15		53964.00
		Total amount for seigniorage charges					406776.00			

AEC
21/2/19
AEC 28/11/19

Deputy Executive Engineer
Water Resources Department
Special Sub Division, Vijayawada

Executive Engineer
Water Resources Department
Special Division, Vijayawada

Office of the Chief Engineer,
Krishna Delta System, Vijayawada.

Proc No: CE/KDS/VJA/OT-4/AEE-11/Drains Krishna/ 368 Dated: 5-02-2019.

Sub:- Water Resources Department - Irrigation - "Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta System" - Technical Sanction - Accorded - Regarding.

Ref:- 1. GO.Rt.No.592 Water Resources (Project-II) Department, Dt:21.08.2018
2. CE /CDO/VJA Lr. No. CE/CDO/EE/Div6/DEE2/KDS/DLRB @KM18.390 on Budameru Dt:1-2-2019
3. SE IC VJA Lr.No.SE/IC/VJA/DB/ATO.2/42^{CE}, Dt:04.02.2019.

Under the circumstances reported by the Superintending Engineer vide reference 3rd cited, the estimate for the work of "Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta System" for an amount of Rs. 354.70 Lakhs **((Rupees Three Crores Fifty Four Lakhs And Seventy Thousand Only))** is technically sanctioned and registered vide CE/KDS R.No: 26 CE/ 2018-19, Dt:5-02-2019 and copy of Technical sanctioned Estimate is herewith communicated for information and taking further necessary action.

The Expenditure is debitable to the head of account "MH 4700-Capital outlay on Major Irrigation - 01 Major Irrigation- Commercial M.H.136 Krishna Delta System G.H.11 State Development Plan S.H. (27) Canals and Distributaries 530 Major works-531 Other Expenditure".

Encl: Copy of Abstract of Tec. Sanctioned Estimate.

To
The Superintending Engineer,
Irrigation Circle, Vijayawada.

Sd/- R.Sathish Kumar 5-2-2019
Chief Engineer
Krishna Delta System
Vijayawada.

//t.c.f//
M. Tulesammet 5/2/19
Officer Technical-4
K.S.P.
5/2/19

ABSTRACT ESTIMATE

Name of work: Construction of Double Lane Road Bridge at Km.18.390 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta Drainage System .

WITH 2018-19 SSR RATES

Sl. No	Quantity	Description of work	Rate	Per	Amount
Bridge Portion					
1	13094.00 Cum	Earthwork in excavation for structures as per drawing and technical specifications Clause 305.1 including setting out, construction of shoring and bracing, removal of stumps and other deleterious material and disposal upto a lead of 50 m, dressing of sides and bottom and backfilling in trenches with excavated suitable material as per Technical Specification 305 MORD / 304 MORTH (Upto 3 M depth)	101.30	1 Cum	1326422
2	6407.00 Cum	Earthwork in excavation for structures as per drawing and technical specifications Clause 305.1 including setting out, construction of shoring and bracing, removal of stumps and other deleterious material and disposal upto a lead of 50 m, dressing of sides and bottom and backfilling in trenches with excavated suitable material as per Technical Specification 305 MORD / 304 MORTH (3 M to 6 M)	116.00	1 Cum	743212
3	496.00 Cum	Filling in foundation trenches with sand as per drawing and technical specification Clause 305.3.9 MORD & 304 MORTH'	514.30	1 Cum	255093
4	600.00 Hours	Dewatering by installing 10HP diesel engine including cost of diesel lubricants, oils, crew and convey engine, with accessories to where even necessary today up the site including to complete the work as directed by the Department	343.00	1 Hour	205800
5	50.00 Cum	Providing and laying insitu vibrated M-10 (28 days cube compressive strength not less than 10 N / sq mm) grade cement concrete using 40 mm down size approved, clean, hard graded aggregates for foundation filling including cost of all materials, machinery, labour, formwork, cleaning, batching, mixing, placing in position, levelling, vibrating, finishing, curing etc., complete with initial lead upto 50 m and all lifts. for Levelling course	5418.00	1 Cum	270900
6	844.00 Cum	Reinforced Cement Concrete M 30 grade for footing raft including reinforcement and formwork complete as per drawing and technical specifications Clauses 800 and 1211 MORD / 1500, 1600, 1700 & 2704 MORTH. C.C. grade M 30 Design Mix, using 20 mm Graded Metal (MORTH 12.8 G) RBR-FNDN-4 for Raft Footing	5381.30	1 Cum	4541817
7	132.00 Cum	R.C.C. grade M 20 (Design Mix) using 20 mm Graded Metal (MORD & MORTH 12.8 C) mechanically mixed, placed in foundation and compacted by vibration including curing for 14 days complete as per drawings for Cutoff (RCC M20) (Design Mix) using 20 mm Graded Metal (MORTH) for Cut off	5128.20	1 Cum	676922

Sl. No	Quantity		Description of work	Rate	Per	Amount
8	47.32	Cum	R.C.C. grade M 25 (Design Mix) using 20 mm Graded Metal (MORD & MORTH 12.8 E) mechanically mixed, placed in foundatuion and compacted by vibration including curing for 14 days complete as per drawings for wings footing	4888.00	1 Cum	231300
9	385.00	Cum	Plain/reinforced cement concrete in substructure complete as per drawings and technical specification Clauses 802, 804, 805, 806, 807, 1202 and 1204 MORD and 1500, 1700 & 2200 MORTH for height upto 5M (piers and abutment) -RCC Grade M25 (Design mix)	5666.00	1 Cum	2181410
10	93.00	Cum	Plain/reinforced cement concrete in substructure complete as per drawings and technical specification Clauses 802, 804, 805, 806, 807, 1202 and 1204 MORD and 1500, 1700 & 2200 MORTH for height from 5M to 10M (piers) -RCC Grade M25 (Design mix)	5835.00	1 Cum	542655
11	970.00	Sqm	Providing and constructing 30 cm thick dry rubble stone pitching including cost of all materials, labour, hand packing, finishing etc., complete(rubble stones : 0.33 cum/sqm)	509.00	1Sqm	493730
12	146810.00	Kgs	Providing, fabricating and placing in position reinforcement steel bars for RCC works including cleaning, straightening, cutting, bending, hooking, lapping, welding wherever required,tying with 1.25 mm dia soft annealed steel wire, including cost of all materials, machinery, labour etc.,complete with initial lead upto 50 and all lifts..	66.00	1 Kgs	9689460
13	107.00	Cum	Providing and laying M-30 grade design mix using 20MM graded metal for reinforced cement concrete in superstructure as per drawing and technical specifications Clauses 800, 1205.4 and 1205.5 MORD / Sections 1500, 1600 & 1700 MORTH for approach slab and kerbs - RCC grade M-30.	5311.00	1 Cum	568277
14	45.00	Cum	Providing and laying cement concrete wearing course M 30 grade including reinforcement complete as per drawing and technical specifications Clauses 800 and 1206.3 MORD and 2702 MORTH for wearing coat	5456.00	1 Cum	245520
15	345.50	Cum	Providing and laying insitu vibrated M-20 (28 days cube compressive strength not less than 20 N / sq mm) grade cement concrete using 20 mm down size approved, clean, hard, graded aggregates for deck siab & kerb including cost of all materials,machinery, labour, formwork,scaffolding, cleaning, batching, mixing, placing in position, levelling, vibrating, finishing, curing etc., complete with initial lead upto 50 m and all lifts for deck slab	9278.50	1 Cum	3205722
16	123.60	Rmt	Construction of R.C.C. railing of M 25 grade in cast-in-situ with 20 mm nominal size aggregate, true to line and grade, tolerance of vertical railing post not to exceed 1 in 500, centre-to-centre spacing between vertical posts not to exceed 2000 mm as per drawing and technical specifications Clauses 800, 900 and 1208.3 MORD. For R.c.C Railing	4582.00	1 Rmt	566335

Sl. No	Quantity		Description of work	Rate	Per	Amount
17	230.67	Nos.	Providing weep holes in brick masonry/stone masonry, plain/reinforced concrete abutment, wing wall, return wall with 100 mm dia AC pipe extending through the full width of the structures with slope of 1(V):20(H) towards drawing face complete as per drawing and technical specification Clauses 614, 709, 1204.3.7 MORD / 2706 & 2200 MORTH.	576.00	1 No.	132866
18	20.00	Nos.	Drainage spouts complete as per drawing and technical specifications Clause 1209 MORD & 2705 MORTH	5791.20	1 No.	115824
19	50.70	Rmt	Providing and fixing in position 20 mm thick premoulded joint filler in expansion joint for fixed ends of simply supported spans not exceeding 10 m to cater for a horizontal movement upto 20 mm, covered with sealant complete as per drawing and technical specifications MORTH	267.00	1 Rmt	13537
20	40.56	Sqm	BMS-W-21 Mastic Pad 2' 0"x4' 0" (0.60M x 1.20 M) of 1" (25.4 MM) thick	839.00	1 Sqm	34030
21	1499.00	Cum	Providing gravel backing and laying filter material underneath pitching in slopes complete as per drawing and technical specifications Clause 1302 MORD / 2504 MORTH	1096.00	1 Cum	1642904
22	787.50	Cum	Construction of granular sub-base by providing coarse graded material, spreading in uniform layers with motor grader on prepared surface, mixing by mix in place method with rotavator at OMC, and compacting with vibratory roller to achieve the desired density, complete as per clause 401 MORTH for diversion road.	1689.00	1 Cum	1330088
23	60	Rmt	Supply of NP3 1000mm dia pipes including collars cost and conveyance of materials to site of work for diversion road.	5426.76	1 Rmt	325606
24	15	Nos.	Cost and supply of RCC collars suitable for 1000MM dia NP-3 clause pipes.	947.00	1 No.	14205
25	15.00	Nos.	Laying and jointing 1000 mm dia. NP-3 class or IRC standard hume pipes in CM 1 : 2 proportion by volume including cost of all materials (excluding pipes and collars), machinery, labour, aligning, packing joints with hemp, finishing, curing etc., complete with initial lead upto 50 m and all lifts for diversion road.	1031.00	1 No.	15465
26	255.00	Cum	Providing, Laying, Spreading and compacting graded HBG crushed stone aggregate to Wet Mix macadem specification including cost, seignorage of all materials and including premixing the material with water at OMC in Mechanical mix plant carriage of mixed material by tipper to site, laying in uniform layers with paver in base courses on well prepared surface and compacting with Vibratory roller to achieve the desired density etc., as directed by the Engineer-in-Charge and as per MoRT&H specification.406 (4th revision) for finished item of work.	1833.00	Cum	467415

Sl. No	Quantity		Description of work	Rate	Per	Amount
27	1275.00	Sqm	Providing and laying surface dressing in single coat using 6 mm nominal size IRC HBG machine crushed stone aggregates @ 0.004 cum / sqm on a layer of bitumen binder of 80/100 grade @ 0.75 kg / sqm including cost, seigniorage and conveyance of all materials to work site and laid on prepared surface and rolling with 8-10 T Power Road Roller etc., complete for finished item of work as per MoRT&H Specification 510 (4th Revision) and as directed by the Engineer-in-charge	41.00	Sqm	52275
28	2550.00	Sqm	Providing and applying tack coat with bitumen Emulsion (Medium setting) (Bulk) using Emulsion pressure distributor at the rate of 0.20 kgs per sqm on the prepared bituminous/granular surface cleaned with mechanical broom for finished item of work as per MoRT&H Specification 503 (4th revision) and as directed by the Engineer-in-Charge.	12.00	Sqm	30600

Sl. No	Quantity	Description of work	Rate	Per	Amount
29	64.00 Cum	Providing and laying of 50mm thick Bituminous Macadam with hot mix plant producing an average output of 37.5 tonnes per hour using hard blasted granite crushed aggregates of Grading - II as per table 500-4 of MoRT&H specification 504 (4th Revision) premixed with bituminous binder of 80/100 grade @ 3.3% by weight of total mixture, transported to site, laid over a previously prepared surface with mechanical paver finisher to the required grade, level and alignment and rolled as per Clauses 501.6 and 501.7 to achieve the desired compaction for finished item of work as directed by the Engineer-in-Charge including hire and operational charges all T&P and all other contingent charges necessary including cost of seigniorage charges on all materials etc., complete and as per MoRTH specification No. 504(4th Revision)	6256.00	Cum	400384
30	32.00 Cum	Providing 25 mm thick compacted Semi-Dense Bituminous Concrete by hot mix plant using hard blasted granite crushed aggregates of Grading -2 as per table 500-15 of specification 508 of MoRT&H (4th Revision), premixed with bituminous binder 60/70 grade @ 5% of mix and filler, transporting the hot mix to work site, laying with mechanical paver finisher to the required grade, level and alignment, rolling with smooth wheeled, vibratory and tandem rollers to achieve the desired compaction as per MoRT&H Specification 508 (4th Revision) complete for finished item of work in all respects as directed by Engineer-in-charge.	7673.00	Cum	245536
31	1417.52 Sqm	Painting to new walls with 2 coats of water proof cement paint (Snowcem) of approved brand and shade over a base coat of approved cement primer grade I making making 3 coats in all to give an even shade after thoroughly brushing the surface to remove all dirt and remains of loose powdered materials, including cost and conveyance of all materials to work site and all operational, incidental, labour charges etc. complete for finished item of work as per SS 912 for internal walls	222.00	Sqm	314689
Working Items Total in Rupees					30879999
Total ECV					30879999
32 L.S	L.S. Items				
33 L.S	Provision towards seigniorage charges				406776
34 L.S	Provision towards GST @ 12%				3705600
35 L.S	Provision towards conducting site survey, drilling of Bore holes and Soil testing and obtaining detailed drawings from CE, CDO				100000
36 L.S	Provision towards S.B.C and conducting plate load test				325000
37 L.S	Provision towards NAC @ 0.1%				30880
	Provision towards photographic charges				21745
Total Amount in Rupees					35470000
(Rupees Three Crores Fifty Four Lakhs and Seventy Thousand only)					
Technical Sanction is hereby accorded for the work of "Construction of Double Lane Road Bridge at Km.18.350 on Budameru drain in Vijayawada city limits in Krishna district under Modernization of Krishna Delta System" for an amount of Rs. 354.70 Lakhs (Rupees Three Crores Fifty Four Lakhs and Seventy Thousand only) and registered vide CE/KDS R.No : 26 CE/2018-19, Dt:5/2/2019. The Expenditure is debitable to the Head of account "MH 4700- Capital outlay on Major Irrigation-01 Major Irrigation- Commercial M.H. 136 Krishna Delta System G.H- State Development Plan SH (27) Canals and Distributaries 530 Major Works-531 other expenditure"					
Sd/-R Satish Kumar, 5-2-2019 Chief Engineer Krishna Delta System Vijayawada					
M. Tulsamane, 5/2/19 Officer Technical-4.					